6.3 Function Fields and the Riemann-Roch Theorem

We develop the theory of "function fields", closely related to the notion of complete (or projective) regular algebraic curves (...).

6.3.1 Relation to the Literature

These treatments fall into two methodological groups:

- Weil's répartition/adele method, in the tradition initiated by Chevalley [Che51]: used by Chevalley himself, by Stichtenoth [Sti09], by Lang [Lan82] (all three over a general constant field), and, at its computational core, even by Serre [Ser59] – whose own bibliographic note attributes the proof to Weil, though he additionally erects the formalism of sheaf cohomology around it, identifying $H^1(X, \mathcal{O}_X)$ with the classes of repartitions. Lang's book, unlike the others in this group, has a different overall aim: a bridge to the classical complex-analytic theory of abelian and theta functions, which the algebraic function-field theory here does not pursue.
- the classical algebraic-geometric (scheme-free, ringed-space) approach shared by Hartshorne [Har13] and Bump [Bum98], the latter specialised to curves with a dedicated treatment of differentials, residues and ramification.

I have interpolated between these, guided by four principles.

Dedekind domains over bare valuation theory. I use Dedekind domain theory for the key results on function fields and the Weil form of Riemann-Roch, rather than the purely valuation-theoretic development of [Che51] and [Sti09]. In the same spirit, finite extensions of $k((t))$ (3.33.4) are analysed directly via the integral closure in a Dedekind domain and a direct power-series computation, rather than via the P-adic completion machinery both of those references build on; this also yields the result in every characteristic, where [Che51]'s analogous Puiseux-expansion treatment (Ch. II, §6) is restricted to the tame case, as noted in the remark opening the Extensions of Power Series Rings section. We also use the theory of affine algebras to simplify the proof of some further technical results.

A general constant field, reduced to the algebraically closed case only once. Riemann-Roch and the Residue Theorem are proved over a general constant field, matching Riemann-Roch (though not the Residue Theorem, which is stated only over an algebraically closed field) in [Sti09], but going beyond [Ser59], [Har13], [Bum98] and [Lan82], whose treatments of Riemann-Roch (and, for [Bum98], the Residue Theorem too) all assume an algebraically closed constant field throughout. The general-field Residue Theorem ((6.3.57)) is not proved directly: it is deduced from the algebraically closed case ((6.3.34)) by base extension to $\overline{k}$ and a Galois-descent argument summing conjugate residues, rather than by redoing the local theory over a general field from scratch.

Concrete constructions over general machinery, wherever one suffices. Like [Che51], [Sti09] and [Lan82], I work with Chevalley and Weil's répartition/adele-style methods directly, and connect this to cohomology only after the fact, rather than building up the general machinery of coherent sheaf cohomology as [Ser59] does around the same underlying répartition computation. Likewise, the Residue Theorem is proved without recourse to local completions, unlike the concrete, differential-level residue theorems of [Ser59] and [Sti09], which both depend on the P-adic completion; for the rational function field this is done via a short, self-contained elementary proof by partial fractions, rather than as a special case of the general machinery. The Trace Formula for Residues ((6.3.42)) likewise avoids completions, via the same Laurent-series finite-extension lemma above, rather than by appeal to the Complete Discrete Valuation Rings machinery developed alongside it (the Extensions of Power Series Rings and Complete Discrete Valuation Rings sections). And residues are defined directly from the uniformiser derivation without ever constructing the module of Kähler differentials $\Omega_{K/k}$, in contrast to [Ser59] and [Sti09], which both identify their (Weil-differential-based) residues with genuine differentials in this sense.

Characteristic p handled uniformly, via universal $\mathbb{Z}$-polynomial identities. Invariance of the residue of a differential under change of uniformiser – notoriously tricky in characteristic p, where the naive derivative-of-a-power argument breaks down whenever p divides the relevant exponent – is settled once, universally, over $\mathbb{Z}[a_1, a_1^{-1}, a_2, a_3, \ldots]$ and specialised ((6.3.26)), the same device used by [Ser59] and [Sti09], rather than the Cartier-operator route of [Bum98] unlike [Ser59] and [Sti09], this is done over a general constant field rather than an algebraically closed one. The same device is reused for the Trace Formula for Residues, which holds without any tameness hypothesis ((6.3.40)) by the identical method [Ser59] uses (there attributed to Hasse) for the analogous local identity $\mathrm{Res}_t(f\,du) = \mathrm{Res}_u(\mathrm{Tr}(f)\,du)$ underlying its own residue formula; it is exactly this universal-identity device that lets the wildly ramified case go through in both results.

Two further points of departure concern scope and connections to the classical picture, rather than proof technique: the dictionary between the function field and the regular algebraic curve – established even over a non-algebraically closed

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field, generalising the algebraically-closed-field statement of [Har13] – is proved later, in (6.2.127) of the Algebraic Geometry chapter's treatment of regular algebraic curves; and, in the classical case of a compact Riemann surface ((6.3.19)), Weil differentials are shown to coincide canonically with classical meromorphic 1-forms, reconnecting the algebraic formalism with the analytic theory it generalises.

6.3.2 Riemann-Roch Theorem

In what follows we generally assume that $\tilde{k} = k$ (i.e. K is geometrically integral).

Definition 6.3.1 (Divisors)
*Let K/k be an algebraic function field. Denote by $\mathrm{Div}(K)$ the free abelian group generated by the places of K/k (**divisor group**). The elements are written as a formal sum*

$$D = \sum_P n_P P$$

*with all but finitely many coefficients $n_P = 0$. Define the **degree** homomorphism*

$$\begin{aligned} \mathrm{Div}(K) &\longrightarrow \mathbb{Z} \\ \sum_P n_P P &\longrightarrow \sum_P n_P \cdot [k(\mathfrak{m}_P) : k] \end{aligned}$$

*We may also define $v_P(D) \coloneqq n_P$, and the **support** of D by $\mathrm{Supp}(D) \coloneqq \{P \mid v_P(D) \neq 0\}$, a finite set. Finally we may define a partial order*

$$D \leq D' \iff v_P(D) \leq v_P(D') \quad \forall P$$

Definition 6.3.2 (Riemann-Roch Space)
*Let K/k be an algebraic function field and D a divisor. Define the **Riemann-Roch Space** associated to D by*

$$L(D) \coloneqq \{x \in K^\star \mid (x) \geq -D\} \cup \{0\}$$

and define

$$\ell(D) \coloneqq \dim_k L(D)$$

(we shortly show it is a k-vector subspace of K)

Definition 6.3.3 (Principal Divisors)
*Let K/k be an algebraic function field and $x \in K^\star$. Then define the **zero divisor** (resp. **pole divisor**, **principal divisor**) of x as follows*

$$\begin{aligned} (x)_0 &\coloneqq \sum_{P : v_P(x) > 0} v_P(x) P, \\ (x)_\infty &\coloneqq \sum_{P : v_P(x) < 0} (-v_P(x)) P, \\ (x) &\coloneqq (x)_0 - (x)_\infty = \sum_P v_P(x) P. \end{aligned}$$

which is well-defined by (6.2.116).

Observe similarly $x \in k^\star \iff (x) = 0$. Define the subgroup of principal divisors by

$$\mathrm{Princ}(K) \coloneqq \{(x) \mid x \in K\}$$

*and the quotient group $\mathrm{Cl}(K) \coloneqq \mathrm{Div}(K)/\mathrm{Princ}(K)$ is called the **divisor class group**. We say two divisors D and D' are **linearly equivalent** if there exists $x \in K$ such that $D - D' = (x)$, equivalently if they become equal in the divisor class group. This is written $D \sim D'$.*

We presently prove that these concepts are well-defined, namely every function $x \in K^\star$ has only finitely many zeros. More precisely we can show

Proposition 6.3.4 (Principal Divisors have degree zero)
Let K/k be an algebraic function field and $x \in K \setminus \tilde{k}$ transcendental. Then

$$\begin{aligned} \deg((x)_0) = \deg((x)_\infty) &= [K : k(x)] \\ \deg((x)) &= 0 \end{aligned}$$

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In particular there is a well-defined homomorphism

$$\deg : \mathrm{Cl}(K) \longrightarrow \mathbb{Z}$$

Proof. Recall (6.2.114) $k[x]$ is a Dedekind domain, $\overline{k[x]}$ is a finite $k[x]$-module and every place $\mathcal{O}_P$ containing x corresponds to prime ideal $\mathfrak{B}$ of $\overline{k[x]}$. By (3.32.15) then $v_P(x)$ is equal to the ramification index $e_{\mathfrak{B}/(x)}$, and the inertial degree is $f_{\mathfrak{B}/(x)} = [k(\mathfrak{m}_P) : k]$ where (x) is a prime ideal of $k[x]$. From (3.35.61) we have the ramification formula

$$[K : k(x)] = \sum_{\mathfrak{B}|(x)} e_{\mathfrak{B}/(x)} f_{\mathfrak{B}/(x)}$$

Therefore $\deg((x)_0) = [K : k(x)]$ and $\deg((x)_\infty) = \deg\left((x^{-1})_0\right) = [K : k(x^{-1})] = [K : k(x)]$. ∎

Proposition 6.3.5

Let K/k be an algebraic function field and D a divisor. Then

a) $L(D)$ is a k-vector subspace of K
b) If $D \sim D'$ then $L(D) = L(D')$
c) $L(0) = \tilde{k}$ so under our standing assumption $\ell(0) = 1$.
d) If $D < 0$ then $L(D) = \{0\}$

Proof.

a) Given $x, y \in L(D)$ then by the strict triangle inequality for valuations $v_P(x+y) \geq \min(v_P(x), v_P(y)) \geq v_P(D)$. Similarly for $\lambda \in k$ we have $v_P(\lambda x) = v_P(\lambda) + v_P(x) = v_P(x)$ (6.2.114).

b) By definition $D' = D + (x)$ we may define a mapping

$$\begin{aligned} L(D) &\longrightarrow L(D') \\ y &\longrightarrow yx \end{aligned}$$

Then

$$v_P(yx) = v_P(x) + v_P(y) \geq v_P(x) + v_P(D) = v_P(D')$$

so this is well-defined. As x^{-1} defines the inverse map it is a bijection.

c) By (6.2.114).a) $\tilde{k} \subset L(0)$. Suppose $x \in K \setminus \tilde{k}$ then by definition x is transcendental and so is x^{-1}. By (6.2.116) x^{-1} has a zero P, i.e. $v_P(x^{-1}) > 0 \implies v_P(x) < 0$. Therefore by definition $x \notin L(0)$ as required.

d) If $x \in \tilde{k}^\star$ then $(x) = 0$ whence $x \notin L(D)$. If $x \in K \setminus \tilde{k}$ then as before x has a pole therefore $x \notin L(D)$ as required.

∎

Lemma 6.3.6

Let K/k be an algebraic function field and $D \leq D'$ divisors. Then $L(D) \subset L(D')$ and

$$\dim_k L(D')/L(D) \leq \deg(D') - \deg(D)$$

In particular $\ell(D)$ is finite and we have

$$\deg(D) - \ell(D) \leq \deg(D') - \ell(D')$$

Proof. Consider a divisor D and suppose $D' := D + P$ for some P possibly in the support of D. Let π be a uniformiser for $\mathcal{O}_P$ and define $t = \pi^{v_P(D')}$. We then have a well-defined map

$$\begin{aligned} \psi : L(D') &\longrightarrow k(\mathfrak{m}_P) \\ x &\longrightarrow xt + \mathfrak{m}_P \end{aligned}$$

Observe that $\psi(x) = 0 \iff xt \in \mathfrak{m}_P \iff v_P(xt) > 0 \iff v_P(x) + v_P(D') > 0 \iff v_P(x) \geq -v_P(D) \iff x \in L(D)$. Therefore $\ker(\psi) = L(D)$ and so

$$\dim_k(L(D')/L(D)) \leq [k(\mathfrak{m}_P) : k] = \deg(D') - \deg(D)$$

Now suppose $D'' = D' + Q$, then by the isomorphism theorem for vector spaces (...)

$$\begin{aligned} \dim_k(L(D'')/L(D)) &= \dim_k(L(D'')/L(D')) + \dim_k(L(D')/L(D)) \\ &\leq \deg(D'') - \deg(D') + \deg(D') - \deg(D) \\ &= \deg(D'') - \deg(D) \end{aligned}$$

from which it follows the general case. ∎

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Proposition 6.3.7 (Dimension Upper Bound)
Let K/k be an algebraic function field and D a divisor. Then

$$\ell(D) \leq \deg(D_+) + \dim_k \tilde{k}$$

where D_+ is the positive part of D.

Proof. $L(D) \subset L(D_+)$ so it is enough to show the result for positive divisors. As $L(0) = \tilde{k}$ (...) the result follows from (6.3.6). ∎

We say that the function $\deg(D) - \ell(D)$ is increasing, we also show it is bounded above.

Lemma 6.3.8 (Dimension Lower Bound I)
Let K/k be an algebraic function field. There exists a positive divisor B and integer $\gamma \in \mathbb{Z}$ such that

$$\deg(mB) - \ell(mB) \leq \gamma$$

for all positive integers m.

Proof. Let $x \in K$ be transcendental and define $B := (x)_\infty$ with $\deg(B) = n$. It's positive by (...). Let $u_1, ..., u_n$ be a $k(x)$-basis for K and a divisor $C \geq 0$ such that $(u_i) \geq -C$. Then we claim that for all integers m the elements $\left\{x^i u_j \;\middle|\; 0 \leq i \leq m, 1 \leq j \leq n\right\}$ are linearly independent over k. But this follows immediately because suppose

$$\sum_{i,j} \lambda_{ij} x^i u_j = 0$$

then

$$\sum_{j=1}^{n} \left(\sum_{i=0}^{m} \lambda_{ij} x^i \right) u_j = 0$$

and by linear independence we have $\lambda_{ij} = 0$.

Further $x^i \in L(iB) \implies x^i \in L(mB)$ and $u_j \in L(C) \implies x^i u_j \in L(mB + C)$. Therefore $\ell(mB + C) \geq n(m+1) = \deg(mB) + n$ for all positive integers m.

On the other hand by (6.3.6) we have $\ell(mB + C) \leq \ell(mB) + \deg(C)$. Therefore

$$\deg(mB) - \ell(mB) \leq \deg(C) - n$$

∎

We may extend this result to all divisors by way of the following lemma.

Lemma 6.3.9
Let K/k be an algebraic function field, B a divisor satisfying the conclusion of (6.3.8) and D an arbitrary divisor. Then there exists divisors D', D'' such that $D \leq D'$, $D' \sim D''$ and $D'' \leq mB$ for some $m \geq 0$.

Proof. Choose a positive divisor $D' \geq D$, then

$$\deg(mB - D') - \ell(mB - D') \leq \deg(mB) - \ell(mB) \leq \gamma$$

for all positive integers m. This means

$$\ell(mB - D') \geq m \deg(B) - \deg(D') - \gamma$$

which is positive for sufficiently large m. Therefore we may choose $0 \neq z \in L(mB - D')$. Let $D'' = D' - (z)$. By definition

$$(z) \geq D' - mB \implies D'' \leq mB$$

as required. ∎

Proposition 6.3.10 (Genus is well-defined)
Let K/k be an algebraic function field. Then there exists a positive integer γ such that

$$\deg(D) - \ell(D) \leq \gamma$$

for all divisors D.

Proof. Then using the notation of (6.3.9) we have

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$$\begin{aligned}\deg(D)-\ell(D) &\overset{(6.3.6)}{\leq} \deg(D')-\ell(D')\\ &\overset{(6.3.4)\,(6.3.5)}{=} \deg(D'')-\ell(D'')\\ &\overset{(6.3.6)}{\leq} \deg(mB)-\ell(mB)\\ &\overset{(6.3.8)}{\leq} \gamma\end{aligned}$$

■

Definition 6.3.11 (Genus of a Function Field)
*Let K/k be an algebraic function field with $\tilde{k}=k$. Define the **genus** by*

$$g := \max\{\deg(D)-\ell(D)+1 \mid D\in \mathrm{Div}(K)\}$$

*Note as $\ell(0)=1$ the genus g is non-negative. For any divisor $D\in\mathrm{Div}(K)$ define the **index of specialty** as follows*

$$i(D) := \ell(D)-\deg(D)+g-1$$

which by definition is a non-negative integer. This satisfies a few key properties (6.3.6)

a) $i(0)=g$
b) $i(D)\geq 0$
c) $D\sim D' \implies i(D)=i(D')$
d) $D'\geq D \implies i(D')\leq i(D)$

The Riemann-Roch theorem then takes the following form

$$\ell(D)-i(D)=\deg(D)-g+1$$

*We say that a divisor D is **non-special** if $i(D)=0$ (equivalently $\ell(D)=\deg(D)-g+1$).*

The full Riemann-Roch amounts to an explicit form of $i(D)$. In order to do this we identify it as the dimension of a certain vector space.

Definition 6.3.12 (Repartitions)
*Let K/k be an algebraic function field. Define the K-algebra of **repartitions** of K by*

$$\mathcal{A}_K := \left\{(\alpha_P)_{P\in\mathbb{P}_K} \;\middle|\; \alpha_P\in K \text{ and } \alpha_P\in\mathcal{O}_P \text{ for all but finitely many places } P\right\}$$

For every divisor $D\in\mathrm{Div}(K)$ define the K-subspace

$$\mathcal{A}_K(D) := \{\alpha\in\mathcal{A}_K \mid v_P(\alpha_P)\geq -v_P(D) \text{ for all } P\in\mathbb{P}_K\}$$

There is a canonical K-algebra homomorphism

$$K\hookrightarrow\mathcal{A}_K$$

which is well-defined by (...). Then clearly we have $K\cap\mathcal{A}_K(D)=L(D)$.

We prove a number of results

Lemma 6.3.13
Let K/k be an algebraic function field. For divisors D, D' such that $D\leq D'$ we have

a) $\dim_k \mathcal{A}_K(D')/\mathcal{A}_K(D)=\deg(D')-\deg(D)$
b) $\dim_k((\mathcal{A}_K(D')+K)/(\mathcal{A}_K(D)+K))=i(D)-i(D')$

Proof. We prove each in turn.

a) By the isomorphism theorem for k-modules it's enough to demonstrate this for the case $D'=D+P$. Let π be a uniformiser for $\mathcal{O}_P$ and let $r:=v_P(D)+1=v_P(D')$. Then define the k-linear map

$$\begin{aligned}\mathcal{A}_K(D') &\longrightarrow k(\mathfrak{m}_P)\\ \alpha &\longrightarrow (\pi^r\alpha_P)(P)\end{aligned}$$

By definition we have $v_P(\alpha_P)\geq -v_P(D')\implies v_P(\pi^r\alpha_P)\geq 0\implies \pi^r\alpha_P\in\mathcal{O}_P$ so that this map is well-defined. Further $\pi^r\alpha_P\in\mathfrak{m}_P\iff v_P(\pi^r\alpha_P)>0\iff v_P(D)+1+v_P(\alpha_P)>0\iff v_P(\alpha_P)\geq -v_P(D)$ whence the kernel is precisely $\mathcal{A}_K(D)$. It is clearly surjective because given $f\in\mathcal{O}_P$ we may consider the repartition given by

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$$(1, ..., 1, \pi^{-r} f, 1, ...)$$

which clearly maps to f under this map. Therefore by the isomorphism theorem we have

$$\dim_k \mathcal{A}_K(D')/\mathcal{A}_K(D) = [k(\mathfrak{m}_P) : k] = \deg(D') - \deg(D)$$

b) We claim there is an exact sequence of k-modules

$$0 \longrightarrow L(D')/L(D) \xrightarrow{\sigma_1} \mathcal{A}_K(D')/\mathcal{A}_K(D) \xrightarrow{\sigma_2} (\mathcal{A}_K(D') + K)/(\mathcal{A}_K(D) + K) \longrightarrow 0$$

from which the result follows immediately by comparing dimensions. The first map is given by

$$\sigma_1([f]) := (v_P(f))_{P \in \mathbb{P}_K}$$

which is obviously well-defined and injective. The map σ_2 is clearly well-defined and surjective. Further it is evident that $\sigma_2 \circ \sigma_1 = 0$. Therefore it only remains to show that $\ker(\sigma_2) \subset \mathrm{im}(\sigma_1)$. Suppose $\alpha \in \mathcal{A}_K(D')$ then $\sigma_2(\alpha) = 0 \implies \alpha - (f) \in \mathcal{A}_K(D) \implies (f) \in \mathcal{A}_K(D')$ for some $f \in K$. Then by definition this means $f \in L(D')$ and $\sigma_1(f) = \alpha$ as required. ∎

Proposition 6.3.14 (Riemann's Theorem)
*Let K/k be an algebraic function field. Suppose D_0 is **non-special** (there always exists such a divisor) then*

a) $\deg(D) > \ell(D_0) \implies D$ *is non-special*
b) $\mathcal{A}_K = \mathcal{A}_K(D_0) + K$

Proof. We prove each in turn

a) For an arbitrary divisor D we have

$$\begin{aligned} \ell(D - D_0) &= i(D - D_0) + \deg(D - D_0) - g + 1 \\ &\geq 0 + \deg(D) - \ell(D_0) \end{aligned}$$

So when $\deg(D) > \ell(D_0)$ we have $\ell(D - D_0) > 0$. Choose $0 \neq f \in L(D - D_0)$. Then we may define $D' := D + (f) \geq D_0$. Then by (...) $i(D) = i(D') \leq i(D_0) = 0$ whence $i(D) = 0$.

b) Given $\alpha \in \mathcal{A}_K$ choose $D \geq D_0$ such that $\alpha \in \mathcal{A}_K(D)$. Then by (6.3.11) $i(D) \leq i(D_0) = 0$ and therefore $i(D) = 0$. By (6.3.13).b) $\mathcal{A}_K(D) + K = \mathcal{A}_K(D_0) + K \implies \alpha \in \mathcal{A}_K(D_0) + K$ as required.

∎

Proposition 6.3.15 (Explicit form of $i(D)$)
Let K/k be an algebraic function field and $D \in \mathrm{Div}(K)$. Then

$$i(D) = \dim_k(\mathcal{A}_K/(\mathcal{A}_K(D) + K))$$

Proof. By (6.3.14) there exists some non-special divisor $D' \geq D$ such that $i(D') = 0$ and $\mathcal{A}_K = \mathcal{A}_K(D') + K$. The formula then follows from (6.3.13).b). ∎

Definition 6.3.16 (Weil Divisors)
Let K/k be an algebraic function field and $D \in \mathrm{Div}(K)$. Define

$$\Omega_K(D) := \{\omega \in \mathrm{Hom}_k(\mathcal{A}_K, k) \mid \mathcal{A}_K(D) + K \subset \ker(\omega)\}$$

*As divisors form a directed system we may form the union of **Weil Divisors***

$$\Omega_K := \bigcup_{D \in \mathrm{Div}(K)} \Omega_K(D) \subset \mathrm{Hom}_k(\mathcal{A}_K, k)$$

Proposition 6.3.17
Let K/k be an algebraic function field. Then there is a canonical isomorphism

$$\mathrm{Hom}_k(\mathcal{A}_K/(\mathcal{A}_K(D) + K), k) \xrightarrow{\sim} \Omega_K(D)$$

In particular $i(D) = \dim_k \Omega_K(D)$. Taking $D = 0$, since $i(0) = g$ (6.3.11) we get

$$\dim_k \Omega_K(0) = g$$

that is, the genus equals the dimension of the space of everywhere-regular Weil differentials.

Proof. This is just the isomorphism theorem for vector spaces. The statement on dimensions follows from (6.3.15) and (3.3.108). ∎

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6.3.3 Meromorphic Differentials in the case $k = \mathbb{C}$

In this section we aim to provide a post-hoc motivation for the introduction of Weil Differentials, by showing that they are canonically isomorphic to the classical space of Meromorphic Differentials in the case $k = \mathbb{C}$. This relies on some techniques which are specific to the complex-analytic structure of Riemann Surfaces.

Proposition 6.3.18 (Residue Theorem for $k = \mathbb{C}$)
Let p be a point of a Riemann surface and π a local holomorphic coordinate centered at p. For f meromorphic near p, define

$$\operatorname{Res}_\pi(f) := \operatorname{Res}_p(f\,\mathrm{d}\pi) := \frac{1}{2\pi i}\oint_\gamma f\,\mathrm{d}\pi$$

for a small loop γ around p – the classical residue of the 1-form $f\,\mathrm{d}\pi$. Writing $g := f \circ \psi$ for the re-expansion of f relative to $\pi' = \psi^{-1}(\pi)$, and $f = \sum_n a_n \pi^n$ for the Laurent series expansion of f at p, then

a) Res_π is k-linear in f;
b) $\operatorname{Res}_\pi(f) = 0$ whenever f is holomorphic at p (i.e. has no pole there);
c) $\operatorname{Res}_\pi(f) = a_{-1}$;
d) $\operatorname{Res}_{\pi'}(g \cdot \psi') = \operatorname{Res}_\pi(f)$;
e) if p ranges over a compact *Riemann surface X and ω is a meromorphic 1-form on X (so has only finitely many poles), then, writing ω_π for the coefficient of ω relative to a local coordinate π at p (so $\omega = \omega_\pi\,\mathrm{d}\pi$ near p, and $\operatorname{Res}_\pi(\omega_\pi) = \operatorname{Res}_p(\omega)$ by definition – independent of the choice of π at each p, by d)),*

$$\sum_p \operatorname{Res}_\pi(\omega_\pi) = 0$$

Proof. Drafted by Claude Sonnet 5, reviewed by David Rufino.

a). Immediate from linearity of the contour integral, $\oint_\gamma (\alpha f + \beta h)\,\mathrm{d}\pi = \alpha \oint_\gamma f\,\mathrm{d}\pi + \beta \oint_\gamma h\,\mathrm{d}\pi$ for $\alpha, \beta \in k$.

b). Immediate from c) below: f holomorphic at p means $a_n = 0$ for every $n < 0$, in particular $a_{-1} = 0$, so $\operatorname{Res}_\pi(f) = a_{-1} = 0$.

c). Term-by-term integration (valid by uniform convergence of the Laurent series on γ) gives $\oint_\gamma f\,\mathrm{d}\pi = \sum_n a_n \oint_\gamma \pi^n\,\mathrm{d}\pi$, and $\oint_\gamma \pi^n\,\mathrm{d}\pi = 2\pi i \delta_{n,-1}$ ($n \neq -1$: exact, $\pi^n\,\mathrm{d}\pi = \mathrm{d}\left(\frac{\pi^{n+1}}{n+1}\right)$, so integrates to 0 around a closed loop; $n = -1$: winding number 1 around $\pi = 0$), so $\operatorname{Res}_\pi(f) = \frac{1}{2\pi i}\oint_\gamma f\,\mathrm{d}\pi = a_{-1}$.

d). By the chain rule, $f\,\mathrm{d}\pi = f(\psi(\pi'))\psi'(\pi')\,\mathrm{d}\pi' = (g \cdot \psi')\,\mathrm{d}\pi'$: these are the same 1-form ω, written in the two coordinates π and π'. The contour integral $\oint_\gamma \omega$ depends only on ω and γ, not on the coordinate used to write it down – reparametrising γ by π' instead of π changes nothing about which loop is being integrated over – so $\operatorname{Res}_p(\omega)$ computed from either expression gives the same number:

$$\operatorname{Res}_{\pi'}(g \cdot \psi') = \operatorname{Res}_p((g \cdot \psi')\,\mathrm{d}\pi') = \operatorname{Res}_p(\omega) = \operatorname{Res}_p(f\,\mathrm{d}\pi) = \operatorname{Res}_\pi(f)$$

e). This uses two facts special to the compact setting, neither purely local (unlike a)–d) above): ω is **closed** away from its poles ($\omega = f\,\mathrm{d}z$ locally with f holomorphic, so $\mathrm{d}\omega = \frac{\partial f}{\partial \overline{z}}\,\mathrm{d}\overline{z} \wedge \mathrm{d}z = 0$ by the Cauchy–Riemann equations), and X is **compact**. Removing a small disk D_p around each pole leaves $X_0 := X \setminus \bigcup_p D_p$, on which ω is holomorphic, so $\mathrm{d}\omega = 0$ there; Stokes' theorem gives

$$0 = \int_{X_0} \mathrm{d}\omega = \int_{\partial X_0} \omega = -\sum_p \oint_{\partial D_p} \omega = -\sum_p 2\pi i \operatorname{Res}_\pi(\omega_\pi)$$

(the sign, as ∂X_0 traverses each ∂D_p oppositely to the usual counterclockwise orientation used to define Res_π, and $\oint_{\partial D_p} \omega = 2\pi i \operatorname{Res}_\pi(\omega_\pi)$ by definition), giving $\sum_p \operatorname{Res}_\pi(\omega_\pi) = 0$. ∎

Proposition 6.3.19 (Classical Differentials are Weil Differentials)
Let X be a compact Riemann surface with function field K (so $k = \mathbb{C}$), and let $\mathcal{M}^1(X)$ denote the $\mathbb{C}$-vector space of meromorphic 1-forms on X. For $\omega \in \mathcal{M}^1(X)$, define $\varphi_\omega \in \operatorname{Hom}_k(\mathcal{A}_K, k)$ by

$$\varphi_\omega(\alpha) := \sum_P \operatorname{Res}_\pi(\alpha_P \cdot \omega_\pi), \quad \alpha = (\alpha_P)_P \in \mathcal{A}_K$$

(a finite sum, by b): $\alpha_P \omega_\pi$ is holomorphic, hence contributes 0, at all but finitely many P). Then $\varphi_\omega \in \Omega_K$, and

$$\mathcal{M}^1(X) \longrightarrow \Omega_K, \quad \omega \mapsto \varphi_\omega$$

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is a canonical $\mathbb{C}$-linear isomorphism.

Proof. Drafted by Claude Sonnet 5, reviewed by David Rufino.

$\varphi_\omega \in \Omega_K$. Let $D := (\omega)$, the divisor of ω (well-defined: $v_P(\omega_\pi)$ is independent of the choice of π at P, as $\omega_{\pi'} = (\omega_\pi \circ \psi) \cdot \psi'$ by the chain rule – the same identity underlying d) – and ψ' is a unit, so composition with ψ and multiplication by ψ' both preserve order of vanishing). We show $\mathcal{A}_K(D) + K \subset \ker(\varphi_\omega)$. For $\alpha \in \mathcal{A}_K(D)$, $v_P(\alpha_P) \geq -v_P(D) = -v_P(\omega_\pi)$ for every P, so $v_P(\alpha_P \omega_\pi) \geq 0$: each $\alpha_P \omega_\pi$ is holomorphic at P, so $\mathrm{Res}_\pi(\alpha_P \omega_\pi) = 0$ by b), giving $\varphi_\omega(\alpha) = 0$. For $f \in K$ (embedded diagonally, $\alpha_P = f$ for every P), $\varphi_\omega(f, f, \ldots) = \sum_P \mathrm{Res}_\pi(f \cdot \omega_\pi) = \sum_P \mathrm{Res}_\pi((f\omega)_\pi) = 0$ by e) (the Residue Theorem), applied to the meromorphic 1-form $f\omega \in \mathcal{M}^1(X)$. So φ_ω vanishes on $\mathcal{A}_K(D) + K$, that is $\varphi_\omega \in \Omega_K(D) \subset \Omega_K$.

$\mathbb{C}$-linearity. Immediate from a): $\varphi_{\lambda\omega_1+\mu\omega_2}(\alpha) = \sum_P \mathrm{Res}_\pi\big(\alpha_P(\lambda\omega_1 + \mu\omega_2)_\pi\big) = \lambda\varphi_{\omega_1}(\alpha) + \mu\varphi_{\omega_2}(\alpha)$ for $\lambda, \mu \in \mathbb{C}$, using that $(\lambda\omega_1 + \mu\omega_2)_\pi = \lambda(\omega_1)_\pi + \mu(\omega_2)_\pi$ (coefficients relative to a fixed π add linearly) and linearity of Res_π in its argument.

Injectivity. Suppose $\varphi_\omega = 0$; we show $\omega = 0$. Fix P and, for $n \in \mathbb{Z}$, let $\alpha^{(n)} \in \mathcal{A}_K$ be the repartition with $\alpha^{(n)}_P := \pi^{-n}$ and $\alpha^{(n)}_Q := 0$ for $Q \neq P$ (a valid repartition: it is holomorphic away from the single place P). Writing $\omega_\pi = \sum_m a_m \pi^m$, $0 = \varphi_\omega(\alpha^{(n)}) = \mathrm{Res}_\pi(\pi^{-n}\omega_\pi) = a_{n-1}$ by c). As n was arbitrary, every coefficient of ω_π vanishes, so ω is 0 near P; as P was arbitrary, $\omega \equiv 0$ on X.

Surjectivity. *Sketch.* This needs the classical Mittag-Leffler theorem for meromorphic differentials: given principal parts (polar Laurent tails) prescribed at finitely many points of X, satisfying the residue-sum-zero constraint of e), there exists a global meromorphic 1-form on X realising them – solved via the $\overline{\partial}$-equation (Hodge theory / finite-dimensionality of $H^1(X, \mathcal{O})$), genuinely global analytic content with no counterpart among a)–e) above. Given $\psi \in \Omega_K(D)$, read off, at each pole P of D, the local principal part ψ demands by testing against the single-place repartitions $\alpha^{(n)}_P = \pi^{-n}$ used in the injectivity argument above; since ψ vanishes on K by construction, these prescribed principal parts automatically satisfy the residue-sum-zero compatibility condition Mittag-Leffler requires, and it produces a global ω realising them, with $\varphi_\omega = \psi$. ∎

6.3.4 Derivations on the Function Field

Proposition 6.3.20 (A Distinguished Basis for $\mathrm{Der}_k(K, K)$)
Let $x \in K$ be transcendental over k and assume k is perfect. Then there is a unique $\partial_x \in \mathrm{Der}_k(K, K)$ with $\partial_x(x) = 1$, and $D = D(x) \cdot \partial_x$ for every $D \in \mathrm{Der}_k(K, K)$.

Proof. Drafted by Claude Sonnet 5, reviewed by David Rufino.

By (3.35.44), $\dim_K \mathrm{Der}_k(K) = 1$. So there is at least one non-zero D_0. If $D_0(x) = 0$ then it vanishes on $k(x)$ by (3.26.9), (3.26.10) and therefore on K as $K/k(x)$ is separable (3.19.103) (3.35.42), a contradiction. Therefore $D_0(x) \neq 0$ for any non-zero derivation D_0. Therefore define $\partial_x := D_0(x)^{-1} D$ and by monogenicity we are done. ∎

Proposition 6.3.21 (Uniformisers are Transcendental)
Let P be a rational place of K/k, π a uniformiser. Then π is transcendental over k.

Proof. By (...) $k \cap \mathfrak{m}_P = (0)$ whence $\pi \notin k$. By assumption $\tilde{k} = k$ whence π is transcendental. ∎

Proposition 6.3.22 (Chain Rule for Uniformiser Derivations)
Let $x, x' \in K$ be transcendental. Then

$$\partial_{x'}(t) = \partial_{x'}(x) \cdot \partial_x(t) \quad \text{for every } t \in K.$$

Proof. This is immediate from (6.3.20). ∎

6.3.5 Residue Symbol at a Rational Place

The residue symbol $\mathrm{Res}_P(z\,\mathrm{d}t)$ built in this subsection is defined via the Laurent coefficients of $z\partial_\pi(t)$ relative to a choice of uniformiser π, and shown independent of that choice (6.3.31) – exactly the invariance a genuine differential-form residue ought to have. This invariance is not straightforward in characteristic p: it reduces (6.3.31) to computing $(\psi^n \psi')_{-1}$ for the change-of-uniformiser series ψ, an identity which the naive derivative-of-a-power argument only settles when $n + 1$ is invertible – automatic in characteristic 0, but not when $p \mid (n + 1)$. We settle it uniformly in every characteristic by proving the identity once, universally, over $\mathbb{Z}[a_1, a_1^{-1}, a_2, a_3, \ldots]$ (indeterminate coefficients, torsion-free over $\mathbb{Z}$) and specialising (6.3.26) – the same universal-$\mathbb{Z}$-polynomial philosophy used, e.g., in [Sti09] (Chap. 4, Prop. 4.2.9); another route entirely avoiding a characteristic-p detour is via finite-potent operators [Tat68] (§§1–2).

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We restrict throughout this subsection to *rational* places P, that is $k(\mathfrak{m}_P) = k$ – geometrically, points of the curve valued in the constant field k itself, rather than in some finite extension of it. This is exactly what allows us to embed K in $k((X))$.

Proposition 6.3.23 (Laurent Coefficients at a Rational Place)
Let P be a rational place of K/k, π a uniformiser of $\mathcal{O}_P$. There is a unique k-linear map

$$L_\pi : K \to k((X))$$

*with $L_\pi(\pi^n) = X^n$ and $v_X(L_\pi(z)) = v_P(z)$ for every $z \in K^\star$ (**L_π preserves valuation**; v_X being the valuation of the DVR $k[[X]]$ (3.10.7)).*

Writing $L_\pi(z) = \sum_j a_j X^j$ for $z \in K^\star$: $a_j = 0$ for $j < v_P(z)$ and $a_{v_P(z)} \neq 0$, and for every $n \geq v_P(z)$,

$$z - \sum_{j=v_P(z)}^{n} a_j \pi^j \in \mathfrak{m}_P^{n+1}$$

*(the **Laurent coefficients** of z at P relative to π). Moreover L_π is injective, $L_\pi(\mathcal{O}_P) \subseteq k[[X]]$ (equivalently, $z \in \mathcal{O}_P \iff L_\pi(z) \in k[[X]]$, for every $z \in K$), and L_π is a k-algebra homomorphism.*

Proof. Drafted by Claude Sonnet 5, reviewed by David Rufino.

As P is rational, $k \to \mathcal{O}_P \to \mathcal{O}_P/\mathfrak{m}_P = k(\mathfrak{m}_P)$ is an isomorphism, that is $k \subseteq \mathcal{O}_P$ is a coefficient field for the DVR $\mathcal{O}_P$. By (3.34.22) (with $A := \mathcal{O}_P$, $\mathfrak{m} := \mathfrak{m}_P$), there is an injective k-algebra homomorphism $L_\pi : K \to k((X))$, extending the injective $L_\pi : \mathcal{O}_P \to k[[X]]$ of that proposition, with $L_\pi(\pi) = X$.

It is valuation-preserving: for $z = a/s \in K^\star$ ($a, s \in \mathcal{O}_P$), $L_\pi(z) = L_\pi(a)L_\pi(s)^{-1}$ (that proposition's extension formula), so $v_X(L_\pi(z)) = v_X(L_\pi(a)) - v_X(L_\pi(s)) = v_P(a) - v_P(s) = v_P(z)$ (multiplicativity of v_X (3.10.6), and the valuation formula on $\mathcal{O}_P$, that proposition again). In particular $z \in \mathcal{O}_P \iff v_P(z) \geq 0 \iff v_X(L_\pi(z)) \geq 0 \iff L_\pi(z) \in k[[X]]$, for $z \in K$.

$L_\pi(\pi^n) = X^n$ for every $n \in \mathbb{Z}$. For $n \geq 0$ this is $L_\pi(\pi)^n = X^n$ (L_π multiplicative); for $n < 0$, as L_π is a field homomorphism and $X = L_\pi(\pi) \neq 0$, $L_\pi(\pi^{-1}) = L_\pi(\pi)^{-1} = X^{-1}$, so $L_\pi(\pi^n) = L_\pi(\pi^{-1})^{-n} = X^n$.

Laurent coefficients. Let $z \in K^\star$, $n_0 := v_P(z)$, and $u := z\pi^{-n_0} \in \mathcal{O}_P$, a unit (as $v_P(u) = 0$, (3.31.2)). By (3.34.22), for every $m \geq 0$ there is a unique $(c_0(u), \ldots, c_m(u)) \in k^{m+1}$ with $u - \sum_{i=0}^{m} c_i(u)\pi^i \in \mathfrak{m}_P^{m+1}$, compatible as m varies, and $c_0(u) \neq 0$ (else $u \in \mathfrak{m}_P$, contradicting u a unit); setting $a_j := c_{j-n_0}(u)$ for $j \geq n_0$ gives $a_{n_0} = c_0(u) \neq 0$, and, multiplying the congruence at $m := n - n_0$ by π^{n_0},

$$z - \sum_{j=n_0}^{n} a_j \pi^j = \pi^{n_0}\left(u - \sum_{i=0}^{n-n_0} c_i(u)\pi^i\right) \in \pi^{n_0}\mathfrak{m}_P^{n-n_0+1} = \mathfrak{m}_P^{n+1}$$

for every $n \geq n_0$ (using $\mathfrak{m}_P = (\pi)$). As $L_\pi(z) = L_\pi(\pi^{n_0})L_\pi(u) = X^{n_0}\sum_i c_i(u)X^i = \sum_{j \geq n_0} a_j X^j$, this is exactly the stated Laurent coefficient sequence for z; uniqueness of $(a_j)_j$ follows from that of $(c_i(u))_i$ (any sequence satisfying the congruence gives, dividing by π^{n_0}, a coefficient sequence for u, forced equal to $(c_i(u))_i$ by (3.34.22)'s uniqueness).

Uniqueness among such maps. Suppose $\phi : K \to k((X))$ is k-linear with $\phi(\pi^n) = X^n$ for every $n \in \mathbb{Z}$ and $v_X(\phi(z)) = v_P(z)$ for every $z \in K^\star$. Fix $z \in K^\star$ with Laurent coefficients $(a_j)_{j \geq v_P(z)}$ as above; for every $n \geq v_P(z)$, writing $z = \sum_{j=v_P(z)}^{n} a_j\pi^j + r_n$ with $r_n \in \mathfrak{m}_P^{n+1}$ (i.e. $r_n = 0$ or $v_P(r_n) \geq n+1$), k-linearity and $\phi(\pi^j) = X^j$ give

$$\phi(z) - \sum_{j=v_P(z)}^{n} a_j X^j = \phi(r_n),$$

with $v_X(\phi(r_n)) \geq n+1$ (either $r_n = 0$, or by hypothesis on ϕ); so $\phi(z) - \sum_{j=v_P(z)}^{n} a_j X^j \in X^{n+1}k[[X]]$, for arbitrarily large n, giving $\phi(z) = \sum_j a_j X^j = L_\pi(z)$. Hence $\phi = L_\pi$. ∎

Definition 6.3.24 (Residue Symbol of a Laurent Series)
Let A be a commutative ring, π an indeterminate, and $z, t \in A((\pi))$. Write ∂_π for the formal derivative, $\partial_\pi\left(\sum_n a_n\pi^n\right) := \sum_n n a_n \pi^{n-1}$, and define

$$\operatorname{Res}_\pi(z\,\mathrm{d}t) := (z\partial_\pi(t))_{-1} \in A,$$

the coefficient of π^{-1} in $z\partial_\pi(t)$.

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Definition 6.3.25 (Residue Symbol at a Rational Place)
Let P be a rational place of K/k, π a uniformiser, and $t, z \in K$. Let $L_\pi : K \hookrightarrow k((\pi))$ be the completion embedding, and define

$$\mathrm{Res}_P(z\,\mathrm{d}t) := \mathrm{Res}_\pi(L_\pi(z)\,\mathrm{d}L_\pi(t)) \in k,$$

where Res_π is the residue of (6.3.24) with $A := k$, and $\mathrm{d}L_\pi(t) := \partial_\pi(L_\pi(t))\,\mathrm{d}\pi$.

This is the special case $A = k$ of (6.3.24), applied to $L_\pi(z), L_\pi(t) \in k((\pi))$. Explicitly, since L_π commutes with the uniformiser derivation, $L_\pi(\partial_\pi(t)) = \partial_\pi(L_\pi(t))$, one has

$$\mathrm{Res}_P(z\,\mathrm{d}t) = L_\pi(z\partial_\pi(t))_{-1} = (L_\pi(z)\partial_\pi(L_\pi(t)))_{-1},$$

which is the formula used previously. We show that it is independent of the choice of uniformiser π below (6.3.31).

Lemma 6.3.26 (Residue of $\psi^n\psi'$)
Let $\psi \in Xk[[X]]$ with $\psi'(0) \neq 0$ (equivalently, ψ has order exactly 1). For every $n \in \mathbb{Z}$,

$$(\psi^n\psi')_{-1} = \delta_{n,-1}$$

Remark 6.3.27
The simplest proof using the identity $(n+1)\psi^n\psi' = (\psi^{n+1})'$ fails when $\mathrm{char}(k) = p \mid n+1$, so some additional work is required.

Proof. Drafted by Claude Sonnet 5, reviewed by David Rufino.

***n* ≥ 0.** $\psi \in Xk[[X]]$ has no negative powers, so neither do ψ^n or ψ'; their product has no X^{-1} term, giving 0, matching $\delta_{n,-1} = 0$.

***n* = −1.** Write $\psi(X) = Xv(X)$ with $v \in k[[X]]^\star$ a unit ($v(0) = \psi'(0) \neq 0$). Then $\psi' = v + Xv'$, so

$$\psi^{-1}\psi' = \frac{v + Xv'}{Xv} = \frac{1}{X} + \frac{v'}{v}$$

The second term is a genuine power series (product of v' and the unit v^{-1}, both in $k[[X]]$), so has zero -1-coefficient; the first contributes $(X^{-1})_{-1} = 1$. Total: 1, matching $\delta_{-1,-1}$.

***n* ≤ −2: a universal identity.** Fix $n \leq -2$; we prove the identity over a universal coefficient ring first, then specialise (see the remark above for why). Let $R := \mathbb{Z}[a_1, a_2, a_3, \ldots][a_1^{-1}]$, the polynomial ring over $\mathbb{Z}$ in countably many indeterminates $a_1, a_2, \ldots$, localised to invert a_1 – an integral domain of characteristic 0 (a localisation of the domain $\mathbb{Z}[a_1, a_2, \ldots]$, itself a domain as a polynomial ring over the domain $\mathbb{Z}$, (3.8.3), applied to finitely many variables at a time since any product only ever involves finitely many of them). Let $\Psi := \sum_{i\geq 1} a_i X^i \in XR[[X]]$, so $\Psi = a_1 X \cdot U$ for $U := 1 + \sum_{i\geq 2}(a_i/a_1)X^{i-1} \in 1 + XR[[X]]$; as U has constant coefficient $1 \in R^\star$, (3.10.2) gives $U \in R[[X]]^\star$, and hence $\Psi \in R((X))^\star$ too ($a_1 \in R^\star$, and X a unit of $R((X))$ directly by construction of the formal Laurent series ring, (3.10.4)). So $\Psi^n\Psi' \in R((X))$ is well-defined, with a well-defined coefficient of X^{-1}, which we still write $(\Psi^n\Psi')_{-1} \in R$.

As before (the power rule and vanishing of a derivative's X^{-1}-coefficient being valid over any commutative ring, not just k), $(n+1)\Psi^n\Psi' = (\Psi^{n+1})'$ ((3.10.10), Ψ being a unit) has vanishing X^{-1}-coefficient:

$$(n+1)\cdot(\Psi^n\Psi')_{-1} = \big((\Psi^{n+1})'\big)_{-1} = 0 \quad \text{in } R$$

As $n \leq -2$, $n+1$ is a nonzero integer; its image in R is therefore nonzero, R having characteristic 0. So $(n+1)\cdot(\Psi^n\Psi')_{-1} = 0$ forces $(\Psi^n\Psi')_{-1} = 0$ in R, by cancellation in the domain R.

Specialising. Write $\psi = \sum_{i\geq 1} c_i X^i$, so $c_1 = \psi'(0) \neq 0$ by hypothesis. As $c_1 \in k^\star$, the assignment $a_1 \mapsto c_1$, $a_i \mapsto c_i$ ($i \geq 2$) extends, by the universal property of localisation (3.6.5) (applied to $\mathbb{Z}[a_1, a_2, \ldots] \to k$, $a_1 \mapsto c_1$ a unit), to a ring homomorphism $\phi : R \to k$. Applying ϕ coefficientwise gives a ring homomorphism $\hat{\phi} : R((X)) \to k((X))$ (directly from the convolution formula for products, (3.10.4)) with $\hat{\phi}(\Psi) = \psi$, compatible with the formal derivative ($\hat{\phi}(f)' = \hat{\phi}(f')$, immediate from the coefficientwise formulas $f' = \sum_m mc_m X^{m-1}$ and $\phi(mc_m) = m\phi(c_m)$) and with the coefficient of X^{-1} (literally the same index, carried along by ϕ). So

$$(\psi^n\psi')_{-1} = \hat{\phi}(\Psi^n\Psi')_{-1} = \phi\big((\Psi^n\Psi')_{-1}\big) = \phi(0) = 0$$

using $(\Psi^n\Psi')_{-1} = 0$ in R (just shown), matching $\delta_{n,-1} = 0$ for $n \leq -2$. ∎

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Lemma 6.3.28 (Laurent Expansions Compose Across Uniformisers)
Let P be a rational place of K/k, π, π' uniformisers at P, and $\psi := L_{\pi'}(\pi) \in Xk[[X]]$.

Then for every $w \in K$,

$$L_{\pi'}(w) = L_{\pi}(w) \circ \psi$$

Proof. Drafted by Claude Sonnet 5, reviewed by David Rufino.

Write $f := L_\pi(w) = \sum_j a_j X^j$ and $g := f \circ \psi = \sum_j a_j \psi^j$ (well-defined, (3.10.13)). As $v(\psi) = 1$ and the order function is multiplicative on the field $k((X))$ (3.10.6), $v(\psi^j) = j$ for every j; so, for every n, the tail $g - \sum_{j\le n} a_j\psi^j = \sum_{j>n} a_j\psi^j$ has every term of order $> n$, that is

$$g - \sum_{j\le n} a_j \psi^j \in X^{n+1}k[[X]] \qquad (\star)$$

For $u \in K$ with $v_P(u) \ge n+1$, (6.3.23) gives $L_{\pi'}(u) \in X^{n+1}k[[X]]$ (its Laurent coefficients vanish below order $v_P(u) \ge n+1$). Apply this to $u := w - \sum_{j\le n} a_j\pi^j \in \mathfrak{m}_P^{n+1}$ (the defining congruence of $f = L_\pi(w)$, (6.3.23)). Since $L_{\pi'}$ is a k-algebra homomorphism of fields (6.3.23) with $L_{\pi'}(\pi) = \psi$, it sends $\pi^j \mapsto \psi^j$ for every $j \in \mathbb{Z}$ (3.2.6), and fixes $a_j \in k$; so, by linearity,

$$L_{\pi'}(w) - \sum_{j\le n} a_j\psi^j = L_{\pi'}\left(w - \sum_{j\le n} a_j\pi^j\right) \in X^{n+1}k[[X]] \qquad (\star\star)$$

Subtracting $(\star)$ from $(\star\star)$, $L_{\pi'}(w) - g \in X^{n+1}k[[X]]$ for every $n \in \mathbb{Z}$; as any nonzero element of $k((X))$ has some fixed finite order, this forces $L_{\pi'}(w) = g = L_\pi(w) \circ \psi$. ∎

Corollary 6.3.29 (Laurent Coefficients of the Uniformiser Derivation)
Let P be a rational place of K/k, π a uniformiser. For every $w \in K$,

$$L_\pi(\partial_\pi(w)) = \frac{\mathrm{d}}{\mathrm{d}X} L_\pi(w)$$

Proof. Drafted by Claude Sonnet 5, reviewed by David Rufino.

Let $M := k((X))$, the target of $L_\pi : K \hookrightarrow k((X))$ (6.3.23), viewed as a K-module through L_π, a k-algebra homomorphism (6.3.23). By (3.35.44) (with $n = 1$, transcendence basis π), $\mathrm{Der}_k(K, M) \cong M$ via $D \mapsto D(\pi)$, so a k-derivation $K \to M$ is uniquely determined by its value at π.

Both $D_1 := L_\pi \circ \partial_\pi$ and $D_2 := \frac{\mathrm{d}}{\mathrm{d}X} \circ L_\pi$ are k-derivations $K \to M$: for D_1, $L_\pi(\partial_\pi(xy)) = L_\pi(x\partial_\pi(y) + y\partial_\pi(x)) = x \cdot D_1(y) + y \cdot D_1(x)$, using additivity and multiplicativity of L_π; for D_2, the same identity holds by the Leibniz rule for formal differentiation together with $L_\pi(xy) = L_\pi(x)L_\pi(y)$; and $D_1(\lambda) = L_\pi(\partial_\pi(\lambda)) = 0 = D_2(\lambda)$ for $\lambda \in k$. Both send $\pi \mapsto 1$ ($L_\pi(\partial_\pi(\pi)) = L_\pi(1) = 1$, $\frac{\mathrm{d}}{\mathrm{d}X}(L_\pi(\pi)) = \frac{\mathrm{d}}{\mathrm{d}X}(X) = 1$), so $D_1 = D_2$, that is $L_\pi(\partial_\pi(w)) = \frac{\mathrm{d}}{\mathrm{d}X} L_\pi(w)$ for every $w \in K$. ∎

Proposition 6.3.30 (The Uniformiser Derivation is Regular)
Let P be a rational place of K/k, π a uniformiser. Then $\partial_\pi(\mathcal{O}_P) \subseteq \mathcal{O}_P$.

Proof. This essentially follows from (6.3.23) and the fact that $\frac{\mathrm{d}}{\mathrm{d}X}$ preserves $k[[X]]$. ∎

Corollary 6.3.31 (Well-Definedness of the Residue Symbol at a Rational Place)
Let P be a rational place of K/k, π, π' uniformisers at P, and $z, t \in K$. Then

$$L_\pi(z\partial_\pi(t))_{-1} = L_{\pi'}(z\partial_{\pi'}(t))_{-1}$$

that is, $\mathrm{Res}_P(z\,\mathrm{d}t)$ of (6.3.25) does not depend on the choice of uniformiser π used to compute it.

Proof. Write $w := z\partial_\pi(t)$ and $\psi := L_{\pi'}(\pi) \in Xk[[X]]$.

$\psi'(0) \neq 0$. Writing $\psi = \sum_j a_j X^j$, this is the Laurent expansion of $\pi \in K^\star$ relative to π' (6.3.23), so $a_j = 0$ for $j < v_P(\pi)$ and $a_{v_P(\pi)} \neq 0$; as $v_P(\pi) = 1$ (π is itself a uniformiser), $a_0 = 0$ and $a_1 \neq 0$. The constant term of $\psi' = \sum_j j a_j X^{j-1}$ is its $j = 1$ term, $1 \cdot a_1 = a_1 \neq 0$, that is $\psi'(0) \neq 0$.

$L_\pi(w)_{-1} = (L_{\pi'}(w) \cdot \psi')_{-1}$. Write $f := L_\pi(w)$ and $g := f \circ \psi$ (well-defined, (3.10.13), as ψ has invertible linear term, just shown). Both sides of $f_{-1} = (g \cdot \psi')_{-1}$ are k-linear in f (composition with ψ, multiplication by ψ', and $(-)_{-1}$ are all k-linear), so it suffices to check $f = X^n$, $n \in \mathbb{Z}$, that is $(\psi^n\psi')_{-1} = \delta_{n,-1}$ – exactly (6.3.26). By (6.3.28), $g = L_\pi(w) \circ \psi = L_{\pi'}(w)$, giving $L_\pi(w)_{-1} = f_{-1} = (g \cdot \psi')_{-1} = (L_{\pi'}(w) \cdot \psi')_{-1}$.

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By (6.3.29) (applied relative to π'), $L_{\pi'}(\partial_{\pi'}(\pi)) = \frac{\mathrm{d}}{\mathrm{d}X} L_{\pi'}(\pi) = \psi'$. As $L_{\pi'}$ is a ring homomorphism (6.3.23),

$$L_\pi(z\partial_\pi(t))_{-1} = L_\pi(w)_{-1} = (L_{\pi'}(w) \cdot L_{\pi'}(\partial_{\pi'}(\pi)))_{-1} = (L_{\pi'}(w\partial_{\pi'}(\pi)))_{-1}.$$

By (6.3.22), $\partial_\pi(t)\partial_{\pi'}(\pi) = \partial_{\pi'}(t)$, so $w\partial_{\pi'}(\pi) = z\partial_\pi(t)\partial_{\pi'}(\pi) = z\partial_{\pi'}(t)$, giving $L_\pi(z\partial_\pi(t))_{-1} = L_{\pi'}(z\partial_{\pi'}(t))_{-1}$. ∎

Lemma 6.3.32 (Residues are Invariant under Isomorphism)
Let $\varphi : K \to K'$ be an isomorphism of k-algebras between algebraic function fields K/k, K'/k, P a rational place of K/k and P' the place of K'/k with $\mathcal{O}_{P'} = \varphi(\mathcal{O}_P)$. Then $\mathrm{Res}_{P'}(\varphi(z)\,\mathrm{d}\varphi(t)) = \mathrm{Res}_P(z\,\mathrm{d}t)$ for all $z, t \in K$.

Proposition 6.3.33 (Elementary Properties of the Residue Symbol at a Rational Place)
Let P be a rational place of K/k, π a uniformiser, $t \in K$. Then

a) $z \mapsto \mathrm{Res}_P(z\,\mathrm{d}t)$ is k-linear $K \to k$;
b) for $x \in K$ transcendental over k and $\partial_x \in \mathrm{Der}_k(K, K)$ the distinguished derivation of (6.3.20), $\mathrm{Res}_P(z\,\mathrm{d}t) = \mathrm{Res}_P((z \cdot \partial_x(t))\,\mathrm{d}x)$ for every $z \in K$;
c) if $z\partial_\pi(t) \in \mathcal{O}_P$ then $\mathrm{Res}_P(z\,\mathrm{d}t) = 0$; in particular this holds if $z, t \in \mathcal{O}_P$.

Proof. Drafted by Claude Sonnet 5, reviewed by David Rufino.

a) By (6.3.25), $\mathrm{Res}_P(z\,\mathrm{d}t) = L_\pi(z\partial_\pi(t))_{-1}$; k-linearity in z follows from that of ∂_π, of L_π (6.3.23), and of coefficient extraction (6.3.24).

b) By (6.3.20) (applied to $D := \partial_\pi$), $\partial_\pi = \partial_\pi(x) \cdot \partial_x$, so $z\partial_\pi(t) = (z \cdot \partial_x(t))\partial_\pi(x)$; applying $L_\pi(-)_{-1}$ to both sides ((6.3.25), once with t and once with x) gives $\mathrm{Res}_P(z\,\mathrm{d}t) = \mathrm{Res}_P((z \cdot \partial_x(t))\,\mathrm{d}x)$.

c) If $z\partial_\pi(t) \in \mathcal{O}_P$ then $L_\pi(z\partial_\pi(t)) \in k[[X]]$ (6.3.23), so its X^{-1}-coefficient vanishes, that is $\mathrm{Res}_P(z\,\mathrm{d}t) = 0$. If $z, t \in \mathcal{O}_P$ then $\partial_\pi(t) \in \mathcal{O}_P$ (6.3.30), so $z\partial_\pi(t) \in \mathcal{O}_P$. ∎

6.3.6 The Algebraic Residue Theorem over an Algebraically Closed Field

Assume throughout this subsection that k is algebraically closed. Then every place of K/k is rational (6.2.114), so the entire theory of (6.3.25) applies directly, with no need for the base-change-to-$\overline{k}$ machinery of the following subsections (which is needed only once k is no longer assumed algebraically closed).

Proposition 6.3.34 (Algebraic Residue Theorem, over an Algebraically Closed Field)
Assume k is algebraically closed. For $t, f \in K$,

$$\sum_P \mathrm{Res}_P(f\,\mathrm{d}t) = 0,$$

summed over all places P of K/k (finitely many terms nonzero, as $f\,\mathrm{d}t$ has only finitely many poles).

Roadmap. The proof, given immediately below, proceeds in the following steps.

- *Reduce to a differential $f\,\mathrm{d}x$.* Choose a separating transcendental $x \in K$; then $f\,\mathrm{d}t = f\partial_x(t)\,\mathrm{d}x$ at every place (6.3.33).b).
- *The case $K = k(x)$.* Partial fractions over $k = \overline{k}$ (6.3.35) write a rational function as a sum of terms $c_{i,j}(x - \theta_i)^{-j}$; the finite residues are the $c_{i,1}$, and the place at infinity has residue $-\sum_i c_{i,1}$ (6.3.39).
- *Trace formula.* For a finite separable $K/k(x)$ and a place Q of $k(x)$, $\sum_{P|Q} \mathrm{Res}_P(f\,\mathrm{d}x) = \mathrm{Res}_Q\big(\mathrm{Tr}_{K/k(x)}(f)\,\mathrm{d}x\big)$ (6.3.42). This is proved locally, after completing at Q: the completed field splits as $\prod_i k((\pi_i))$ over the places above Q, reducing the identity to the local trace formula at each place, which is computed monomial by monomial via the roots of unity in k, following Lang.
- *Assemble.* Group the places P of K by the place Q of $k(x)$ below them. The trace formula turns each inner sum into $\mathrm{Res}_Q\big(\mathrm{Tr}_{K/k(x)}(f')\,\mathrm{d}x\big)$, and summing over Q gives 0 by the case $K = k(x)$.

Proof of (6.3.34). Drafted by Claude Sonnet 5, reviewed by David Rufino.

Fix any transcendental $x \in K$ with $K/k(x)$ finite and separable (exists as K/k is a function field, separably generated), and let $\partial_x \in \mathrm{Der}_k(K, K)$ be the distinguished element of (6.3.20) with $\partial_x(x) = 1$. By (6.3.33).b), for every place P,

$$\mathrm{Res}_P(f\,\mathrm{d}t) = \mathrm{Res}_P((f \cdot \partial_x(t))\,\mathrm{d}x)$$

this handles $t \in k$ automatically too, as $\partial_x(k) = 0$ then makes both sides 0, matching $\mathrm{d}t = 0$. So it suffices to prove $\sum_P \mathrm{Res}_P(f'\,\mathrm{d}x) = 0$ for $f' := f \cdot \partial_x(t) \in K$.

Every place P of K/k lies over a unique place Q of $k(x)/k$, so (finitely many terms being nonzero) the sum may be regrouped:

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$$\sum_{P \text{ of } K/k} \operatorname{Res}_P(f'\,\mathrm{d}x) = \sum_{Q \text{ of } k(x)/k} \ \sum_{P \mid Q} \operatorname{Res}_P(f'\,\mathrm{d}x).$$

By (6.3.42), the inner sum is $\operatorname{Res}_Q\big(\operatorname{Tr}_{K/k(x)}(f')\,\mathrm{d}x\big)$ for every Q, so

$$\sum_{P} \operatorname{Res}_P(f'\,\mathrm{d}x) = \sum_{Q} \operatorname{Res}_Q\big(\operatorname{Tr}_{K/k(x)}(f')\,\mathrm{d}x\big) = 0$$

by (6.3.39), applied to $\operatorname{Tr}_{K/k(x)}(f') \in k(x)$. ∎

This proof uses only (6.3.20) (already available, with no need for the module of differentials $\Omega_{K/k}$ to have been formally constructed).

Lemma 6.3.35 (Partial Fraction Decomposition over an Algebraically Closed Field)
Let k be algebraically closed and $r, s \in k[X]$ polynomials with $s \neq 0$. Write $s(X) = c\prod_{i=1}^{m}(X-\theta_i)^{e_i}$ for distinct $\theta_i \in k$, integers $e_i \geq 1$, and $c \in k^\times$. Then there exist a polynomial $Q \in k[X]$ and constants $c_{i,j} \in k$ $(1 \leq i \leq m,\ 1 \leq j \leq e_i)$ with

$$\frac{r(X)}{s(X)} = Q(X) + \sum_{i=1}^{m}\sum_{j=1}^{e_i} \frac{c_{i,j}}{(X-\theta_i)^j}$$

as an identity in $k(X)$. If $\deg r < \deg s$ then $Q = 0$.

Proof. Absorbing c into r, we may take $c = 1$. We argue by induction on m, the number of distinct roots of s. If $m = 0$ then $s = 1$ and we take $Q := r$.

Let $m \geq 1$ and set $s_1 = (X-\theta_1)^{e_1}$ and $s_2(X) := \prod_{i=2}^{m}(X-\theta_i)^{e_i}$, so $s = s_1 s_2$. By (3.19.10) s_1 and s_2 are co-prime as they have no common roots in k, so $gs_1 + hs_2 = 1$ for some $g, h \in k[X]$ and

$$\frac{r(X)}{s(X)} = \frac{r(X)h(X)}{s_1}(X) + \frac{r(X)g(X)}{s_2}(X)$$

By the division algorithm $rh = Q_1 s_1 + r_1$ with $\deg(r_1) < e_1$. By (3.8.11) $r_1 = \sum_{l=0}^{e_1-1} d_l (X-\theta_1)^l$ for some $d_l \in k$, so with $c_{1,j} := d_{e_1-j}$

$$\frac{rh}{s_1} = Q_1 + \sum_{j=1}^{e_1} \frac{c_{1,j}}{(X-\theta_1)^j}$$

The induction hypothesis applies to rg/s_2, which has $m-1$ distinct roots in its denominator, and gives the terms with $i \geq 2$ together with a polynomial Q_2. Adding, the claim holds with $Q := Q_1 + Q_2$.

Finally suppose $\deg r < \deg s$. Multiplying the decomposition by s gives $Qs = r - \sum_{i,j} c_{i,j}\, s/(X-\theta_i)^j$, where each $s/(X-\theta_i)^j$ is a polynomial of degree $\deg s - j \leq \deg s - 1$. So the right hand side has degree less than $\deg s$, whereas $\deg(Qs) \geq \deg s$ if $Q \neq 0$; hence $Q = 0$. ∎

Remark 6.3.36 (A Complex-Analytic Heuristic)
Over $k = \mathbb{C}$, this also follows from Liouville's theorem. At each θ_i, choose $c_{i,1}, \ldots, c_{i,e_i}$ to match the principal part of the Laurent expansion of r/s at θ_i (possible and unique, exactly as the induction above extracts them); then $r/s - \sum_{i,j} c_{i,j}/(X-\theta_i)^j$ is a rational function with no poles anywhere in $\mathbb{C}$, hence an entire function, and one that grows at most polynomially at infinity (being itself a rational function) – so by (the polynomial-growth form of) Liouville's theorem it is a polynomial, namely Q.

Lemma 6.3.37 (Residue at Infinity of a Proper Rational Function)
Let $f, g \in k[X]$ with $g \neq 0 \neq f$ and $\deg(f) < \deg(g)$. Then, for the place at infinity of $k(X)/k$ (6.2.117),

$$\operatorname{Res}_\infty\left(\frac{f}{g}\,\mathrm{d}X\right) = \begin{cases} -\frac{\ell(f)}{\ell(g)} & \deg(f) = \deg(g) - 1 \\ 0 & \deg(f) < \deg(g) - 1. \end{cases}$$

Proof. Drafted by Claude Sonnet 5, reviewed by David Rufino.

Let $p := \deg(f)$, $q := \deg(g)$, and $m := q - p \geq 1$. Set $A(U) := U^p f(1/U)$, $B(U) := U^q g(1/U) \in k[X]$, polynomials with $A(0) = \ell(f)$, $B(0) = \ell(g) \neq 0$, so $w := A/B \in k[[U]]$ is a unit with $w(0) = \ell(f)/\ell(g)$. As $f(1/U) = U^{-p}A(U)$ and $g(1/U) = U^{-q}B(U)$,

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$$\frac{f}{g}(1/U) = U^{q-p}\frac{A(U)}{B(U)} = U^m w(U).$$

Let U be an indeterminate and $\varphi : k(X) \to k(U)$ the isomorphism with $\varphi(X) = U^{-1}$; it maps the place at infinity to the place 0 of $k(U)$, with uniformiser U (6.2.117). By (6.3.32) and (6.3.33).b) (with $x := U$, as $\partial_U(U^{-1}) = -U^{-2}$),

$$\mathrm{Res}_\infty\left(\frac{f}{g}\,\mathrm{d}X\right) = \mathrm{Res}_0(\varphi(f/g)\,\mathrm{d}(U^{-1})) = \mathrm{Res}_0(-U^{m-2}w(U)\,\mathrm{d}U).$$

As $\partial_U(U) = 1$, this is the coefficient of U^{-1} in $-U^{m-2}w(U)$, that is $-$ the coefficient of U^{1-m} in $w(U)$. If $m = 1$ this is $-w(0) = -\ell(f)/\ell(g)$. If $m \geq 2$ then $1 - m < 0$, and as $w \in k[[U]]$ has only non-negative powers, this coefficient is 0. ∎

Remark 6.3.38 (A Complex-Analytic Heuristic)
Over $k = \mathbb{C}$, this is controlled entirely by the leading-order behaviour of f/g at infinity. Writing $U = 1/X$, the residue at infinity is $-\mathrm{Res}_0$ of the pulled-back differential, and $f/g(1/U)\cdot\partial_U(U^{-1}) = -U^{m-2}w(U)$ has leading term $-\left(\frac{\ell(f)}{\ell(g)}\right)U^{m-2}$ ($w(0) = \ell(f)/\ell(g) \neq 0$); classically $\oint U^n\,\mathrm{d}U = 0$ for every $n \neq -1$, so the residue vanishes unless this leading *term is itself the U^{-1} term, i.e. $m - 2 = -1$, or $\deg f = \deg g - 1$ – exactly the borderline decay rate $f/g \sim (\ell(f)/\ell(g))X^{-1}$ at infinity – in which case the residue is read off directly from the leading coefficients, with no correction from the rest of the expansion of w.*

Lemma 6.3.39 (Residue Theorem for the Rational Function Field)
Assume k is algebraically closed. For $\theta \in k$, $n \in \mathbb{Z}$ and $g := (X-\theta)^n \in k(X)$, the residues at the places of $k(X)/k$ (the $\mathcal{O}_{X-\theta'}$, $\theta' \in k$, written θ', and the place at infinity (6.2.117)) are

$$\mathrm{Res}_{\theta'}(g\,\mathrm{d}X) = \begin{cases} 1 & \theta' = \theta \text{ and } n = -1 \\ 0 & \text{otherwise} \end{cases} \qquad \mathrm{Res}_\infty(g\,\mathrm{d}X) = \begin{cases} -1 & n = -1 \\ 0 & n \neq -1. \end{cases}$$

Consequently, for every $g \in k(X)$, where $k(X)/k$ is the rational function field,

$$\sum_Q \mathrm{Res}_Q(g\,\mathrm{d}X) = 0,$$

summed over all places Q of $k(X)/k$ (finitely many terms nonzero).

Proof. Drafted by Claude Sonnet 5, reviewed by David Rufino.

Write T for the variable of the Laurent series field, to avoid a clash with $X \in k(X)$, so that $L_\pi : k(X) \to k((T))$ is a ring homomorphism with $L_\pi(\pi) = T$ (6.3.23).

Finite places. The place θ' has uniformiser $\pi := X - \theta'$ with $\partial_\pi(X) = 1$. If $\theta' \neq \theta$ then g and X lie in $\mathcal{O}_{X-\theta'}$, so $\mathrm{Res}_{\theta'}(g\,\mathrm{d}X) = 0$ (6.3.33).c). If $\theta' = \theta$ then by (6.3.25)

$$\mathrm{Res}_\theta(g\,\mathrm{d}X) = L_\pi(\pi^n)_{-1} = (T^n)_{-1} = \begin{cases} 1 & n = -1 \\ 0 & n \neq -1 \end{cases}$$

The place at infinity. If $n \geq 0$ then g is a polynomial, so $g(1/U)$ is a k-linear combination of $U^0, \ldots, U^{-n}$, and $g(1/U)\partial_U(X) = -U^{-2}g(1/U)$ a linear combination of $U^{-2}, \ldots, U^{-n-2}$; as every exponent is ≤ -2, $\mathrm{Res}_\infty(g\,\mathrm{d}X) = 0$. If $n \leq -1$ then $g = 1/(X-\theta)^{-n}$ is a proper fraction with numerator 1 and denominator $(X-\theta)^{-n}$ of degree $-n$, so by (6.3.37)

$$\mathrm{Res}_\infty(g\,\mathrm{d}X) = \begin{cases} -1 & n = -1 \\ 0 & n \leq -2. \end{cases}$$

The sum vanishes for every $g \in k(X)$. By (6.3.35) every $g \in k(X)$ is the sum of a polynomial and a k-linear combination of the functions $(X-\theta)^{-n}$ with $\theta \in k$, $n \geq 1$; and by (3.8.11) every polynomial is a k-linear combination of the $(X-\theta)^n$, $n \geq 0$, for any fixed $\theta \in k$. The residue being k-linear, this reduces to the case $g = (X-\theta)^n$ above, where the residues at all places other than θ and ∞ vanish, and $\mathrm{Res}_\theta(g\,\mathrm{d}X) + \mathrm{Res}_\infty(g\,\mathrm{d}X)$ is $1 - 1 = 0$ if $n = -1$ and 0 otherwise. ∎

Lemma 6.3.40 (Local Trace Formula for Residues)
Let k be algebraically closed, $F := k((t))$, and let $E = k((\pi))$ be a finite separable extension of F with $v_\pi(t) = e$, so $[E:F] = e$ and $t = \pi^e w$ for some $w \in k[[\pi]]^\star$ (3.33.4). For $g \in E$ write $\mathrm{Res}_\pi(g\,\mathrm{d}t) := (g\partial_\pi(t))_{-1}$. Then, in every characteristic,

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$$\operatorname{Res}_t\big(\operatorname{Tr}_{E/F}(g)\,\mathrm{d}t\big) = \operatorname{Res}_\pi(g\,\mathrm{d}t) \qquad (g \in E).$$

Proof. Drafted by Claude Sonnet 5, reviewed by David Rufino.

Basis and matrices. By (3.33.2) (with $R := k$, $X := \pi$, $m := e$, $T = t$), every element of $k[[\pi]]$ is uniquely $\sum_{i<e} f_i(t)\pi^i$ with $f_i \in k[[Y]]$; multiplying by powers of t (which is π^e up to a unit) gives the same for E. So $1, \pi, \dots, \pi^{e-1}$ is an F-basis of E. For $g \in E$ write $g\pi^i = \sum_{j<e} M_{ji}(g)\pi^j$ with $M_{ji}(g) \in F$, and $\operatorname{Tr}_{E/F}(g) = \sum_{i<e} M_{ii}(g)$.

Universal model. Let $R := \mathbb{Z}\big[b_0^{\pm 1}, b_1, b_2, \dots\big]$, a domain of characteristic 0, and put $w_R := \sum_{i\ge 0} b_i\pi^i \in R[[\pi]]^\star$, $t_R := \pi^e w_R$. The construction above works verbatim over R (only b_0 needs inverting), with matrix entries $M_{ji}(g) \in R((t_R))$ given by universal formulas in $+$, $\times$ and w_R^{-1}, whose coefficients lie in R. For $r \in \mathbb{Z}$ set

$$\Psi(\pi^r) := \operatorname{Res}_{t_R}(\operatorname{Tr}_R(\pi^r)\,\mathrm{d}t_R) - \operatorname{Res}_\pi(\pi^r\,\mathrm{d}t_R) \in R,$$

both terms being coefficients of elements of $R((\pi))$ or $R((t_R))$.

Claim: $\Psi(\pi^r) = 0$ for all r. Let Ω be an algebraic closure of $\operatorname{Frac}(R)$; it has characteristic 0, so e is invertible in Ω and $R \subset \Omega$. Write $w_R = b_0 u$ with $u \in 1 + \pi R[[\pi]]$. The binomial series $\varepsilon := \sum_{i\ge 0}\binom{1/e}{i}(u-1)^i \in \Omega[[\pi]]$ satisfies $\varepsilon^e = u$, and for $\beta \in \Omega$ with $\beta^e = b_0$ the element $\varpi := \beta\pi\varepsilon$ satisfies $\varpi^e = t_R$. As $\varpi = \pi \cdot (\text{unit})$, it is a uniformiser of $\Omega((\pi)) = \Omega((\varpi))$, and $\Omega((t_R)) \subset \Omega((\pi))$ has basis $1, \varpi, \dots, \varpi^{e-1}$ (the argument above, with ϖ in place of π).

Take $g = \varpi^r$ with $r = es + j$, $0 \le j < e$. Then $\varpi^r = t_R^s\varpi^j$, and multiplication by ϖ^j sends the basis vector ϖ^i to ϖ^{i+j}, which is a basis vector times t_R^0 or t_R. So the matrix of ϖ^r has nonzero entries only in positions shifted by r modulo e: its diagonal vanishes unless $e \mid r$, and if $r = es$ it is the scalar t_R^s. Hence

$$\operatorname{Tr}(\varpi^r) = \begin{cases} e t_R^s & r = es \\ 0 & e \nmid r. \end{cases}$$

Therefore $\operatorname{Res}_{t_R}(\operatorname{Tr}(\varpi^r)\,\mathrm{d}t_R) = e$ if $r = -e$ and 0 otherwise. On the other side $\mathrm{d}t_R = e\varpi^{e-1}\,\mathrm{d}\varpi$, so $\varpi^r\,\mathrm{d}t_R = e\varpi^{r+e-1}\,\mathrm{d}\varpi$ and $\operatorname{Res}_\varpi(\varpi^r\,\mathrm{d}t_R)$ is again e for $r = -e$ and 0 otherwise. Residues are independent of the choice of uniformiser (6.3.31), so $\operatorname{Res}_\varpi(g\,\mathrm{d}t_R) = \operatorname{Res}_\pi(g\,\mathrm{d}t_R)$. Thus the identity holds for $g = \varpi^r$ for every r. Both sides are Ω-linear and continuous in g, and the ϖ^r span $\Omega((\pi))$ topologically, so it holds for every $g \in \Omega((\pi))$; in particular $\Psi(\pi^r) \in R$ maps to 0 in Ω. As $R \subset \Omega$, $\Psi(\pi^r) = 0$.

Specialisation. Given $E = k((\pi))$ and $t = \pi^e w$ with $w = \sum_{i\ge 0} c_i\pi^i$, $c_0 \in k^\times$, let $\phi : R \to k$ be the ring homomorphism with $b_i \mapsto c_i$ (it exists by the universal property of localisation, as c_0 is a unit). Applying ϕ coefficientwise gives ring homomorphisms $R((\pi)) \to k((\pi))$ and $R((t_R)) \to k((t))$ (with $t_R \mapsto t$, since $\phi(w_R) = w$). These commute with the formation of matrices M_{ji}, hence with traces, and with the coefficient extraction defining both residues. So $\phi(\Psi(\pi^r))$ is the defect $\operatorname{Res}_t\big(\operatorname{Tr}_{E/F}(\pi^r)\,\mathrm{d}t\big) - \operatorname{Res}_\pi(\pi^r\,\mathrm{d}t)$ over k, and it equals 0. Finally, for $g = \sum_r a_r\pi^r \in k((\pi))$ both sides depend on only finitely many a_r, since $\operatorname{Tr}(\pi^r)$ has t-order tending to ∞ with r and only finitely many terms contribute to a given coefficient of t^{-1} or π^{-1}. The identity for each monomial therefore gives it for g. ∎

Lemma 6.3.41 (Places over P and Irreducible Factors)
Let k be algebraically closed, $K/k(x)$ a finite separable extension of degree n, P the place (x) of $k(x)$, $K_P := k((x))$, and $y \in K$ a primitive element with minimal polynomial g over $k(x)$. Factor $g = g_1 \cdots g_r$ into distinct monic irreducibles over K_P, with $d_j := \deg g_j$, and let $E_j := K_P[Y]/(g_j)$ with $\iota_j : K \hookrightarrow E_j$ sending $y \mapsto Y$. Then $j \mapsto Q_j$, defined below, is a bijection from $\{1, \dots, r\}$ onto the places of K/k over P, and $e_{Q_j} = d_j$.

Proof. Drafted by Claude Sonnet 5, reviewed by David Rufino.

The place Q_j. E_j/K_P is finite separable (as g_j is separable), so by (3.33.4) there is an isomorphism $\Psi_j : E_j \xrightarrow{\sim} k\big((\pi_j)\big)$ over K_P with $\Psi_j(x)$ of order d_j. Pulling back the valuation of $k\big((\pi_j)\big)$ along $\Psi_j \circ \iota_j$ gives a discrete valuation of K extending v_P with ramification index d_j; let Q_j be the corresponding place.

Every place is some Q_j. Let Q be a place of K/k over P. As k is algebraically closed, Q is rational, so (6.3.23) gives a valuation-preserving embedding $L_Q : K \hookrightarrow k\big((\pi_Q)\big)$. Since $g(y) = 0$, the element $L_Q(y)$ is a root of g in $k\big((\pi_Q)\big)$. The g_j are pairwise coprime in $K_P[Y]$, so $L_Q(y)$ is a root of exactly one g_j (a common root of $g_j, g_{j'}$ would be a root of 1, by Bézout). Call this $j = j(Q)$. Then L_Q factors through E_j, and $L_Q(x)$ has ramification index e_Q over K_P; the valuation induced on E_j extends v_P, so by the uniqueness of extension ((3.33.5), applied to E_j/K_P) it equals the valuation of Q_j. Hence $Q = Q_{j(Q)}$.

Injectivity. If $Q_j = Q_{j'}$ then $j(Q_j) = j$ and $j(Q_{j'}) = j'$ are determined by the same valuation, and the minimal factor of $L(y)$ is unique, so $j = j'$. ∎

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Proposition 6.3.42 (Trace Formula for Residues)
Assume k is algebraically closed. Let $\mathcal{O} \coloneqq k[x]$, $\mathcal{B} \coloneqq \overline{\mathcal{O}}^K$ the integral closure of $\mathcal{O}$ in a finite separable extension $K/k(x)$, P a place of $k(x)/k$ corresponding to a maximal ideal $\mathfrak{p} \lhd \mathcal{O}$, and $f \in K$. Then

$$\sum_{Q|P} \operatorname{Res}_Q(f\,\mathrm{d}x) = \operatorname{Res}_P\big(\operatorname{Tr}_{K/k(x)}(f)\,\mathrm{d}x\big),$$

with no hypothesis on $\operatorname{char}(k)$.

Proof. Drafted by Claude Sonnet 5, reviewed by David Rufino.

As k is algebraically closed every place is rational, so $\mathfrak{p} = (x-\theta)$ for some $\theta \in k$; replacing x by $x-\theta$ we may assume $\mathfrak{p} = (x)$.

By (3.32.15) the places $Q_1, \dots, Q_r$ of K/k over P have ramification indices $e_1, \dots, e_r$ with $\sum_i e_i = n \coloneqq [K : k(x)]$ (residue degree 1, as k is algebraically closed). Let $K_P \coloneqq k((x))$, with $k(x) \hookrightarrow K_P$ the expansion at P (6.3.23).

Factoring the minimal polynomial. Let $y \in \mathcal{B}$ be a primitive element with minimal polynomial $g \in \mathcal{O}[Y]$, monic of degree n. Factor $g = g_1 \cdots g_r$ over K_P as in (6.3.41), with $E_j = K_P[Y]/(g_j)$, $\iota_j : K \hookrightarrow E_j$, and $\Psi_j : E_j \xrightarrow{\sim} k\big((\pi_j)\big)$ the isomorphisms given there, so that $\Psi_j(x)$ has order $d_j = \deg g_j$. Put $\lambda_j \coloneqq \Psi_j \circ \iota_j : K \hookrightarrow k\big((\pi_j)\big)$ and $L_{\pi_j} \coloneqq \lambda_j$. By (6.3.41), the places Q_j attached to the g_j are exactly the places of K/k over P, with $e_{Q_j} = d_j$, so after relabelling Q_i is the place attached to g_i and $e_i = d_i$.

Trace additivity. By the Chinese remainder theorem applied to $g = \prod_j g_j$,

$$K \otimes_{k(x)} K_P \xrightarrow{\sim} \prod_{i=1}^{r} E_i, \quad f \otimes 1 \mapsto (\iota_i(f))_i,$$

and composing with the isomorphisms Ψ_i identifies the right side with $\prod_i k((\pi_i))$, with $f \otimes 1 \mapsto \big(L_{\pi_i}(f)\big)_i$. The trace is unchanged by base change and additive over a finite product, so

$$\operatorname{Tr}_{K/k(x)}(f) = \sum_{i=1}^{r} \operatorname{Tr}_{k((\pi_i))/K_P}\big(L_{\pi_i}(f)\big) \qquad (f \in K).$$

Local identities. The residue at Q_i is the residue symbol of (6.3.25), computed in $k((\pi_i))$ through L_{π_i}, so $\operatorname{Res}_{Q_i}(f\,\mathrm{d}x) = \operatorname{Res}_{\pi_i}\big(L_{\pi_i}(f)\,\mathrm{d}x\big)$ (Laurent residue of (6.3.24)), and $\operatorname{Res}_P(z\,\mathrm{d}x) = \operatorname{Res}_x(z\,\mathrm{d}x)$ for $z \in K_P = k((x))$. Applying (6.3.40) to $E = k((\pi_i))$ over $F = K_P$ (with $t = x$ and $\Psi_i(x) = \tau_i$) gives, for every $f \in K$,

$$\operatorname{Res}_{Q_i}(f\,\mathrm{d}x) = \operatorname{Res}_P\big(\operatorname{Tr}_{k((\pi_i))/K_P}\big(L_{\pi_i}(f)\big)\,\mathrm{d}x\big).$$

Assembly. Summing over i and using k-linearity of Res_P and trace additivity,

$$\sum_{Q|P} \operatorname{Res}_Q(f\,\mathrm{d}x) = \sum_{i=1}^{r} \operatorname{Res}_P\big(\operatorname{Tr}_{k((\pi_i))/K_P}\big(L_{\pi_i}(f)\big)\,\mathrm{d}x\big) = \operatorname{Res}_P\big(\operatorname{Tr}_{K/k(x)}(f)\,\mathrm{d}x\big).$$

■

6.3.7 Base Extension of the Constant Field

Throughout this section, in addition to the standing hypothesis that K/k is geometrically integral ($\tilde{k} = k$), assume also that k is **perfect**. Let k'/k be an algebraic field extension, and write $K_{k'} \coloneqq K \otimes_k k'$.

Proposition 6.3.43 (The Base-Changed Function Field)
$K_{k'}$ is a field, and an algebraic function field over k', geometrically integral; if k'/k is finite, $[K_{k'} : K] = [k' : k]$.

Proof. Drafted by Claude Sonnet 5, reviewed by David Rufino.

$K_{k'}$ is a field. As K/k is geometrically integral and k'/k is algebraic, $K_{k'} = K \otimes_k k'$ is a field (3.36.45).

$K_{k'}/k'$ is geometrically integral. By associativity of tensor products (3.4.13) and $\overline{k'} = \overline{k}$ (k'/k algebraic), $K_{k'} \otimes_{k'} \overline{k'} \cong K \otimes_k \big(k' \otimes_{k'} \overline{k}\big) = K \otimes_k \overline{k}$, a field, as K/k is geometrically integral.

$K_{k'}/k'$ is an algebraic function field. Fix a transcendental $x \in K$ with $K/k(x)$ finite (exists, as K/k is an algebraic function field). As $k[x] \otimes_k k' \cong k'[x]$ (tensoring a polynomial ring with a field extension), the image of x in $K_{k'}$ remains transcendental over k'. The image of $k(x) \otimes_k k'$ in $K_{k'}$ (well-defined and injective, as $k(x) \subset K$ is a k-subalgebra, (3.36.2)) is then a subring of the field $K_{k'}$, hence a domain; as k'/k is algebraic, it is thus a field (3.35.57),

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and – being the subfield of $K_{k'}$ generated by k' and the transcendental x – equals $k'(x)$. By associativity of tensor products (3.4.13) (as K is a $k(x)$-module),

$$K_{k'} = K \otimes_k k' \cong K \otimes_{k(x)} (k(x) \otimes_k k') = K \otimes_{k(x)} k'(x)$$

As $K/k(x)$ is finite, K is a finite-dimensional $k(x)$-vector space, hence free (3.3.125) with a basis of size $n := [K : k(x)]$; extension of scalars (3.4.26) then makes $K_{k'} \cong K \otimes_{k(x)} k'(x)$ a free $k'(x)$-module with a basis of the same size n, in particular finite over $k'(x)$. So $K_{k'}$ is a finitely generated field extension of k' (generated by x together with a finite $k'(x)$-basis) of transcendence degree 1: an algebraic function field.

Degree. If k'/k is finite, k' is a free k-module with a basis of size $[k' : k]$, so extension of scalars along $k \hookrightarrow K$ (3.4.26) makes $K_{k'} = K \otimes_k k'$ a free K-module with a basis of the same size, giving $[K_{k'} : K] = [k' : k]$. ∎

Proposition 6.3.44 (Constant Field Extension is Unramified)
Let P be a place of K/k and k'/k algebraic. Every place P' of $K_{k'}/k'$ lying over P is unramified.

There are exactly r such places, where r is the number of maximal ideals of $A := k(\mathfrak{m}_P) \otimes_k k'$ (finite, even when k'/k is not). Moreover each such P' satisfies $\mathcal{O}_{P'} \cap K = \mathcal{O}_P$ and $\mathfrak{m}_{P'} \cap K = \mathfrak{m}_P$.

More precisely, writing $\mathcal{O}_{P,k'} := \mathcal{O}_P \otimes_k k'$, $A \cong \mathcal{O}_{P,k'}/\mathfrak{m}_P\mathcal{O}_{P,k'}$ as k'-algebras (3.4.42) there is a bijection

$$\begin{aligned} \operatorname{Specm}(k(\mathfrak{m}_P) \otimes_k k') &\longleftrightarrow \{\text{places of } K_{k'}/k' \text{ lying over } P\} \\ \mathfrak{p}'/\mathfrak{m}_P\mathcal{O}_{P,k'} &\longrightarrow \left(\mathcal{O}_{P,k'}\right)_{\mathfrak{p}'} \\ \left(\mathfrak{m}_{P'} \cap \mathcal{O}_{P,k'}\right)/\mathfrak{m}_P\mathcal{O}_{P,k'} &\longleftarrow \mathcal{O}_{P'} \end{aligned}$$

Proof. Drafted by Claude Sonnet 5, reviewed by David Rufino.

We claim that we are in the situation of (3.36.47), applied with $\mathcal{O} := \mathcal{O}_P$, $F := K$:

- $\mathcal{O}_P$ is a discrete valuation ring containing k (a place of K/k is by definition a valuation ring $k \subsetneq \mathcal{O}_P \subsetneq K$ (6.2.112), automatically discrete (6.2.114))
- with fraction field K which is geometrically integral (the standing hypothesis of this section)
- $k(\mathfrak{m}_P)$ is finite over k (6.2.114)
- every valuation ring R of $K_{k'}/k'$ with $k' \subset R$ and $\mathcal{O}_P \subset R$ satisfies $R \cap K = \mathcal{O}_P$: such an R is exactly a place of $K_{k'}/k'$, and this is (6.2.119) (applied with $L := K_{k'}$, $k'' := k'$)

Therefore the result follows from there: writing $\mathcal{O}_{P,k'} := \mathcal{O}_P \otimes_k k'$, its $\mathfrak{b} := \mathfrak{m}_P\mathcal{O}_{P,k'}$ has quotient $\mathcal{O}_{P,k'}/\mathfrak{b} \cong A$ (above), so a) gives exactly r maximal ideals $\mathfrak{p}_1, \ldots, \mathfrak{p}_r$ of $\mathcal{O}_{P,k'}$; by b), each $\left(\mathcal{O}_{P,k'}\right)_{\mathfrak{p}_j}$ is a place P'_j of $K_{k'}/k'$ (with $\mathcal{O}_{P'_j} := \left(\mathcal{O}_{P,k'}\right)_{\mathfrak{p}_j}$), unramified over $\mathcal{O}_P$, and, as in the statement, corresponds to $\mathfrak{p}_j/\mathfrak{b} \in \operatorname{Specm}(A)$; and, by c) together with the last point above, *every* place P' of $K_{k'}/k'$ over P has $\mathcal{O}_{P'} = \left(\mathcal{O}_{P,k'}\right)_{\mathfrak{p}_j}$ for some j, that is $P' = P'_j$, giving the stated bijection. As $K_{k'}/k'$ is itself an algebraic function field (6.3.43), every place of $K_{k'}/k'$ – in particular P'_j – is automatically a discrete valuation ring (6.2.114), so $e(P'_j/P) = v_{P'_j}(\pi_P) = 1$: unramified, with π_P a uniformiser of $\mathcal{O}_P$ (3.31.2). Finally $\mathcal{O}_{P'_j} \cap K = \mathcal{O}_P$ and $\mathfrak{m}_{P'_j} \cap K = \mathfrak{m}_P$ by e). ∎

Lemma 6.3.45 (A Rational Place Extends Uniquely Under Any Algebraic Base Change)
Let P be a rational place of K/k. Then for any algebraic k'/k, there is a unique place Q of $K_{k'}$ over P, and $e(Q/P) = 1$, with Q again rational over k'.

Proof. Drafted by Claude Sonnet 5, reviewed by David Rufino.

This is the case $k(\mathfrak{m}_P) = k$ of (6.3.44). ∎

Proposition 6.3.46 (Complete Splitting)
Let K/k be an algebraic function field, P a place of K/k, and k'/k an algebraic extension. Suppose further that k' contains a normal closure of $k(\mathfrak{m}_P)$.

*Then P **splits completely** in $K_{k'}/k'$: that is there are exactly $n = [k(\mathfrak{m}_P) : k]$ places above P, each rational over k'.*

This holds in particular when $k' = \overline{k}$.

Proof. Drafted by Claude Sonnet 5, reviewed by David Rufino.

Write $k_P := k(\mathfrak{m}_P)$; k_P/k is finite (6.2.114) and separable, as k is perfect (3.19.103). As k' contains a normal closure of k_P/k, (3.36.15) gives $k_P \otimes_k k' \cong (k')^n$ as k'-algebras, with exactly n maximal ideals, each with residue field k'

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(the coordinate kernels). By (6.3.44) (with $A := k_P \otimes_k k'$) this implies that P splits completely into n places, each rational (residue field k'). ∎

Proposition 6.3.47 (Automorphisms Act Transitively on Places over $\overline{k}$)
Let P be a place of K/k, $d := [k(\mathfrak{m}_P) : k]$, and $P'_1, ..., P'_d$ the places of $K_{\overline{k}}/\overline{k}$ over P ((6.3.46), with $k' := \overline{k}$). For $\sigma \in \mathrm{Aut}\left(\overline{k}/k\right)$, the map

$$\sigma_* : K_{\overline{k}} \to K_{\overline{k}}, \qquad x \otimes c \mapsto x \otimes \sigma(c)$$

is a well-defined ring automorphism of $K_{\overline{k}}$ fixing K pointwise, and $\sigma \mapsto \sigma_*$ is a group action of $\mathrm{Aut}\left(\overline{k}/k\right)$ on $K_{\overline{k}}$.

For any place P' of $K_{\overline{k}}/\overline{k}$ lying over P then $\sigma_*(\mathcal{O}_{P'})$ is a place lying over P and

$$v_{\sigma(P')}(\sigma_*(z)) = v_{P'}(z)$$

for all $z \in K_{\overline{k}}^\star$; this defines a transitive group action of $\mathrm{Aut}\left(\overline{k}/k\right)$ on $\{P'_1, ..., P'_d\}$.

Moreover, for P' among $P'_1, ..., P'_d$ (rational over $\overline{k}$), write $\iota_{P'} : k(\mathfrak{m}_P) \hookrightarrow k(\mathfrak{m}_{P'}) = \overline{k}$ for the residue embedding induced by $\mathcal{O}_P \subset \mathcal{O}_{P'}$ (well-defined as $\mathfrak{m}_P \subset \mathfrak{m}_{P'} \cap \mathcal{O}_P$). Then $\iota_{\sigma(P')} = \sigma \circ \iota_{P'}$.

This yields a bijection

$$\begin{aligned} \{\text{places of } \ K/k\} &\longleftrightarrow \left\{\mathrm{Aut}\left(\overline{k}/k\right)\text{-orbits of places of } \ K_{\overline{k}}/\overline{k}\right\} \\ P &\longrightarrow \{P'_1, ..., P'_d\} \end{aligned}$$

Proof. As $\sigma : \overline{k} \to \overline{k}$ is a k-algebra homomorphism, $1_K \otimes \sigma$ is a well-defined k-algebra endomorphism of $K \otimes_k \overline{k} = K_{\overline{k}}$ (3.4.36).

As $(1_K \otimes \sigma) \circ (1_K \otimes \tau) = 1_K \otimes (\sigma\tau)$ then σ_* is an automorphism of $K_{\overline{k}}/K$ and $\sigma \mapsto \sigma_*$ is a group homomorphism.

As σ_* is a field automorphism of $K_{\overline{k}}$ then $\sigma_*(\mathcal{O}_{P'})$ is a place of $K_{\overline{k}}$ which lies over the same place as $\mathcal{O}_{P'}$, as σ_* fixes $K \supset \mathcal{O}_P$.

The valuation identity follows because σ_* restricts to an isomorphism on the valuation rings.

As k is perfect, $k(\mathfrak{m}_P)/k$ is separable (3.19.103), so (3.36.15) gives a canonical isomorphism of $\overline{k}$-algebras

$$k(\mathfrak{m}_P) \otimes_k \overline{k} \cong \overline{k}^d, \quad p(\theta) \otimes c \mapsto (c\,p(\theta_1), ..., c\,p(\theta_d)) \qquad \left(p \in k[X], c \in \overline{k}\right)$$

identifying the d maximal ideals of this fibre with $\mathfrak{p}_j$, the kernel of the j-th coordinate.

Let $\mathfrak{p}'_1, ..., \mathfrak{p}'_d$ denote the corresponding maximal ideals of $\mathcal{O}_{P,\overline{k}} = \mathcal{O}_P \otimes_k \overline{k}$. Define $\mathcal{O}_{P'_i} := \left(\mathcal{O}_{P,\overline{k}}\right)_{\mathfrak{p}'_i}$ then by (6.3.44) these are precisely the places $P'_1, ..., P'_d$ above P, and $\mathcal{O}_{P'_i} \cap K = \mathcal{O}_P$, which shows that the given map is injective.

By (3.19.66) $\mathrm{Aut}\left(\overline{k}/k\right)$ permutes the roots θ_i transitively. We may show that if $\sigma(\theta_j) = \theta_{j'}$ then $\sigma_*\left(\mathcal{O}_{P'_j}\right) = \mathcal{O}_{P'_{j'}}$ and the action on places is transitive and $\{P'_1, ..., P'_d\}$ is precisely an $\mathrm{Aut}\left(\overline{k}/k\right)$-orbit.

Given a place $\overline{P}$ of $K_{\overline{k}}/\overline{k}$, $\mathcal{O} := \mathcal{O}_{\overline{P}} \cap K$ is a (proper) valuation ring of K by (6.2.118), giving a place P of K/k with $\mathcal{O}_P = \mathcal{O} \subset \mathcal{O}_{\overline{P}}$, that is $\overline{P}$ lies over P. As $P'_1, ..., P'_d$ are all the places of $K_{\overline{k}}/\overline{k}$ above P (above), $\overline{P}$ is one of them. So the given map is a surjection. ∎

Lemma 6.3.48 (Residue Embeddings are Exactly the Embeddings into $\overline{k}$)
Let P be a place of K/k, $d := [k(\mathfrak{m}_P) : k]$, and $P'_1, ..., P'_d$ the places of $K_{\overline{k}}/\overline{k}$ over P ((6.3.46), with $k' := \overline{k}$).

Denote by $\iota_j : k(\mathfrak{m}_P) \hookrightarrow k\left(\mathfrak{m}_{P'_j}\right) = \overline{k}$ the corresponding embeddings, then these exhaust all the k-algebra embeddings $k(\mathfrak{m}_P)/k \hookrightarrow \overline{k}/k$.

Proof. By (6.3.47) there are $d = [k(\mathfrak{m}_P) : k]$ places and, as k is perfect so $k(\mathfrak{m}_P)/k$ is separable (3.19.103), there are $\#\,\mathrm{Mor}_k\left(k(\mathfrak{m}_P), \overline{k}\right) = [k(\mathfrak{m}_P) : k]_s = d$ k-algebra embeddings (3.19.83).

We show that $j \mapsto \iota_j$ is injective, i.e. that $\iota_j = \iota_{j'}$ implies $\mathcal{O}_{P'_j} = \mathcal{O}_{P'_{j'}}$ (and so $j = j'$). Consider the composite $\rho_j : \mathcal{O}_{P,\overline{k}} \to \left(\mathcal{O}_{P,\overline{k}}\right)_{\mathfrak{p}'_j} = \mathcal{O}_{P'_j} \to \mathcal{O}_{P'_j}/\mathfrak{m}_{P'_j} = \overline{k}$. It is $\overline{k}$-linear and is uniquely determined by

$$\rho_j(a \otimes c) = c\,\iota_j(\overline{a}) \qquad \left(a \in \mathcal{O}_P, c \in \overline{k}\right),$$

so $\iota_j = \iota_{j'} \Longrightarrow \rho_j = \rho_{j'}$, and their kernels agree, that is $\mathfrak{p}'_j = \mathfrak{p}'_{j'}$ (as $\ker \rho_j = \mathfrak{m}_{P'_j} \cap \mathcal{O}_{P,\overline{k}} = \left(\mathfrak{p}'_j\left(\mathcal{O}_{P,\overline{k}}\right)_{\mathfrak{p}'_j}\right)^c = \mathfrak{p}'_j$ (3.6.42)). By (6.3.44) $P'_j = P'_{j'}$ and we are done by counting. ∎

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6.3.8 Residue Symbol at a General Place

At a non-rational place there is no element of k to reduce against: (6.3.23)'s recursion needs, at each step, a canonical way to lift a class of $k(\mathfrak{m}_P)$ back into $\mathcal{O}_P$, and $k(\mathfrak{m}_P) \neq k$ means $k \subset \mathcal{O}_P$ no longer supplies one. Rather than search for a section of $\mathcal{O}_P \twoheadrightarrow k(\mathfrak{m}_P)$ directly, we base change to $\overline{k}$ instead: P splits completely into $d := [k(\mathfrak{m}_P) : k]$ *rational* places of $K_{\overline{k}}/\overline{k}$, each of which already has the Laurent coefficient theory of (6.3.23) available. The d resulting elements of $\overline{k}$ are exactly the Galois conjugates of a single element of $k(\mathfrak{m}_P)$, which we take as the residue at P.

Definition 6.3.49 (Conjugate Places)
Let P be a place of K/k, $d := [k(\mathfrak{m}_P) : k]$, and $\iota_1, \dots, \iota_d : k(\mathfrak{m}_P) \hookrightarrow \overline{k}$ the embeddings, with P'_j the corresponding place of $K_{\overline{k}}/\overline{k}$ over P, $\iota_j = \iota_{P'_j}$ (6.3.48). As each P'_j/P is unramified, every uniformiser π at P remains a uniformiser at P'_j (6.3.44), so the residue symbol $\mathrm{Res}_{P'_j}(w\,\mathrm{d}s)$ at the rational place P'_j of $K_{\overline{k}}/\overline{k}$ (6.3.25) is defined for $w, s \in K_{\overline{k}}$.

Lemma 6.3.50 (The Uniformiser Derivation Extends to the Base Change)
Let P be a place of K/k, π a uniformiser, and $P'_1, \dots, P'_d$ the conjugate rational places of $K_{\overline{k}}/\overline{k}$ over P (6.3.49), at each of which $\pi \otimes 1$ remains a uniformiser.

Then $\partial_{\pi\otimes 1} \in \mathrm{Der}_{\overline{k}}(K_{\overline{k}}, K_{\overline{k}})$ is the unique D satisfying

$$D|_K = \partial_\pi$$

and is in particular regular at every place P'_j.

Proposition 6.3.51 (Conjugate Residues are $\mathrm{Aut}\left(\overline{k}/k\right)$-Equivariant)
For $\sigma \in \mathrm{Aut}\left(\overline{k}/k\right)$, $j = 1, \dots, d$, writing j' for the index with $\sigma\left(P'_j\right) = P'_{j'}$ (6.3.47), and $t, z \in K$,

$$\sigma\Big(\mathrm{Res}_{P'_j}((z\otimes 1)\,\mathrm{d}(t\otimes 1))\Big) = \mathrm{Res}_{P'_{j'}}((z\otimes 1)\,\mathrm{d}(t\otimes 1))$$

Proof. Drafted by Claude Sonnet 5, reviewed by David Rufino.

Fix a uniformiser π at P, and write $w := z\,\partial_\pi(t) \in K$; by (6.3.50), $\mathrm{Res}_{P'_j}((z\otimes 1)\,\mathrm{d}(t\otimes 1)) = L_\pi(w\otimes 1)_{-1}$ (and likewise at $P'_{j'}$). Write σ_* for the induced automorphism of $K_{\overline{k}}$ (6.3.47), fixing K pointwise, and c_i for the i-th Laurent coefficient of $w \otimes 1$ at P'_j relative to π, so for every n,

$$w \otimes 1 - \sum_{i \le n} c_i \pi^i \in \mathfrak{m}_{P'_j}^{n+1}$$

Applying σ_* (a ring automorphism fixing $w \otimes 1$ and π, both in K) and using $\sigma_*\left(\mathfrak{m}_{P'_j}^{n+1}\right) = \mathfrak{m}_{P'_{j'}}^{n+1}$ (as σ_* restricts to an isomorphism $\mathcal{O}_{P'_j} \xrightarrow{\sim} \mathcal{O}_{P'_{j'}}$ (6.3.47)),

$$w \otimes 1 - \sum_{i \le n} \sigma(c_i) \pi^i \in \mathfrak{m}_{P'_{j'}}^{n+1}$$

for every n. By uniqueness of Laurent coefficients ((6.3.23), at $P'_{j'}$), $(\sigma(c_i))_i$ are exactly the Laurent coefficients of $w \otimes 1$ at $P'_{j'}$ relative to π; taking $i = -1$, $\sigma(L_\pi(w\otimes 1)_{-1}) = \sigma(c_{-1}) = L_\pi(w\otimes 1)_{-1}$ at $P'_{j'}$, that is $\sigma\Big(\mathrm{Res}_{P'_j}((z\otimes 1)\,\mathrm{d}(t\otimes 1))\Big) = \mathrm{Res}_{P'_{j'}}((z\otimes 1)\,\mathrm{d}(t\otimes 1))$. ∎

Proposition 6.3.52 (Conjugate Residue Sum is a Trace)
For $t, z \in K$, there is a single $\alpha \in k(\mathfrak{m}_P)$ with $\iota_j(\alpha) = \mathrm{Res}_{P'_j}((z\otimes 1)\,\mathrm{d}(t\otimes 1))$ for every $j = 1, \dots, d$ (well-defined, see proof), and

$$\sum_{j=1}^{d} \mathrm{Res}_{P'_j}((z\otimes 1)\,\mathrm{d}(t\otimes 1)) = \mathrm{Tr}_{k(\mathfrak{m}_P)/k}(\alpha) \in k$$

Proof. Drafted by Claude Sonnet 5, reviewed by David Rufino.

Write $R_j := \mathrm{Res}_{P'_j}((z\otimes 1)\,\mathrm{d}(t\otimes 1))$ for brevity.

$R_1 \in \iota_1(k(\mathfrak{m}_P))$. Write $F := \iota_1(k(\mathfrak{m}_P)) \subset \overline{k}$. As $\overline{k}/k$ is Galois (k perfect) with subfield F, $\overline{k}/F$ is itself Galois (3.19.116), with $\mathrm{Gal}\left(\overline{k}/F\right) = \mathrm{Aut}\left(\overline{k}/F\right)$ (3.19.51). For $\sigma \in \mathrm{Aut}\left(\overline{k}/F\right)$, σ fixes $F = \iota_1(k(\mathfrak{m}_P))$ pointwise, that is $\sigma \circ \iota_1 = \iota_1$, so $\sigma(P'_1) = P'_1$ (6.3.47), and (6.3.51) gives $\sigma(R_1) = R_1$: that is $R_1 \in \overline{k}^{\mathrm{Gal}(\overline{k}/F)}$. So (3.19.117) gives $\overline{k}^{\mathrm{Gal}(\overline{k}/F)} = F$, that is $R_1 \in F = \iota_1(k(\mathfrak{m}_P))$; write $\alpha := \iota_1^{-1}(R_1) \in k(\mathfrak{m}_P)$.

$\iota_j(\alpha) = R_j$ for every j. By transitivity (6.3.47), choose $\sigma \in \mathrm{Aut}\left(\overline{k}/k\right)$ with $\sigma(P'_1) = P'_j$, so $\sigma \circ \iota_1 = \iota_j$ (6.3.47). By equivariance (6.3.51), $\iota_j(\alpha) = (\sigma \circ \iota_1)(\alpha) = \sigma(\iota_1(\alpha)) = \sigma(R_1) = R_j$, the single element promised in the statement.

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Conclusion. By (6.3.48), $\mathrm{Mor}_k\big(k(\mathfrak{m}_P), \overline{k}\big) = \{\iota_1, \dots, \iota_d\}$; as k is perfect, $k(\mathfrak{m}_P)/k$ is separable (3.19.103), so $[k(\mathfrak{m}_P) : k]_i = 1$ (3.19.93). So (3.19.129) (trace is sum of embeddings in $\overline{k}$) gives

$$\mathrm{Tr}_{k(\mathfrak{m}_P)/k}(\alpha) = \sum_{j=1}^{d} \iota_j(\alpha) = \sum_{j=1}^{d} R_j$$

In particular $\sum_{j=1}^{d} R_j \in k$, $\mathrm{Tr}_{k(\mathfrak{m}_P)/k}$ taking values in k by definition (3.19.122). ∎

Definition 6.3.53 (Residue Symbol at a General Place)
Let P be a place of K/k, and $t, z \in K$. Define

$$\mathrm{Res}_P(z\,\mathrm{d}t) := \sum_{j=1}^{d} \mathrm{Res}_{P_j'}((z \otimes 1)\,\mathrm{d}(t \otimes 1)) \in k$$

(residues at the conjugate rational places $P_1', \dots, P_d'$, (6.3.49)) – landing in k by (6.3.52).

Once the module of differentials $\Omega_{K/k}$ is available, it is a 1-dimensional K-vector space dual to $\mathrm{Der}_k(K, K)$ (both being $\mathrm{trdeg}_k(K) = 1$-dimensional, (3.35.44)), and the pairing of $z\,\mathrm{d}t \in \Omega_{K/k}$ with ∂_π is exactly $z\,\partial_\pi(t)$; $\mathrm{Res}_P(z\,\mathrm{d}t)$ above is then the residue, in this sense, of the differential $z\,\mathrm{d}t$.

Proposition 6.3.54 (Elementary Properties of the Residue Symbol at a General Place)
Let P be a place of K/k, $t \in K$. Then

a) $z \mapsto \mathrm{Res}_P(z\,\mathrm{d}t)$ is k-linear $K \to k$;
b) if P is rational (so $d = 1$, P_1' canonically identified with P), $\mathrm{Res}_P(z\,\mathrm{d}t)$ agrees with (6.3.25), for every $z \in K$.

Proof. Drafted by Claude Sonnet 5, reviewed by David Rufino.

***k*-linearity.** Fix a uniformiser π at P. For $z, z' \in K$ and $a, b \in k$, $(az + bz')\partial_\pi(t) = a(z\partial_\pi(t)) + b(z'\partial_\pi(t))$ (K-bilinearity of multiplication, k-linearity of ∂_π), so by (6.3.50) and k-linearity of $L_\pi(-)_{-1}$ ((6.3.23) and (6.3.24)),

$$\mathrm{Res}_{P_j'}(((az + bz') \otimes 1)\,\mathrm{d}(t \otimes 1)) = a\,\mathrm{Res}_{P_j'}((z \otimes 1)\,\mathrm{d}(t \otimes 1)) + b\,\mathrm{Res}_{P_j'}((z' \otimes 1)\,\mathrm{d}(t \otimes 1))$$

for every j; summing over j gives $\mathrm{Res}_P((az + bz')\,\mathrm{d}t) = a\,\mathrm{Res}_P(z\,\mathrm{d}t) + b\,\mathrm{Res}_P(z'\,\mathrm{d}t)$.

Rational case. If P is rational, $d = 1$, and P_1' is identified with P under $K \subset K_{\overline{k}}$ (with any uniformiser π at P remaining one at P_1'); the sum defining $\mathrm{Res}_P(z\,\mathrm{d}t)$ then has the single term $\mathrm{Res}_{P_1'}((z \otimes 1)\,\mathrm{d}(t \otimes 1)) = \mathrm{Res}_P(z\,\mathrm{d}t)$ of (6.3.25) applied verbatim (as $z \otimes 1, t \otimes 1$ agree with z, t under $K \subset K_{\overline{k}}$, and P_1' with P). ∎

Lemma 6.3.55 (Regular Differentials have Zero Residue)
Let P be a place of K/k, π a uniformiser at P, and $t, z \in K$ with $z\,\partial_\pi(t) \in \mathcal{O}_P$. Then $\mathrm{Res}_P(z\,\mathrm{d}t) = 0$. In particular this holds if P is rational and $z, t \in \mathcal{O}_P$.

Proof. Drafted by Claude Sonnet 5, reviewed by David Rufino.

Write $w := z\,\partial_\pi(t) \in \mathcal{O}_P$. By (6.3.50), for every j,

$$\mathrm{Res}_{P_j'}((z \otimes 1)\,\mathrm{d}(t \otimes 1)) = L_\pi(w \otimes 1)_{-1}.$$

As P_j' lies over P, $w \otimes 1 \in \mathcal{O}_P \otimes 1 \subseteq \mathcal{O}_{P_j'}$, so $L_\pi(w \otimes 1)$ has no negative powers (6.3.23) and its X^{-1}-coefficient is 0. Summing over j gives $\mathrm{Res}_P(z\,\mathrm{d}t) = 0$. If P is rational and $t \in \mathcal{O}_P$ then $\partial_\pi(t) \in \mathcal{O}_P$ (6.3.30), so $z\,\partial_\pi(t) \in \mathcal{O}_P$ when also $z \in \mathcal{O}_P$. ∎

Proposition 6.3.56 (Change of Base Differential)
Let P be a place of K/k, $x \in K$ transcendental over k, and $\partial_x \in \mathrm{Der}_k(K, K)$ the distinguished derivation of (6.3.20). For $t, z \in K$,

$$\mathrm{Res}_P(z\,\mathrm{d}t) = \mathrm{Res}_P((z \cdot \partial_x(t))\,\mathrm{d}x).$$

Proof. Drafted by Claude Sonnet 5, reviewed by David Rufino.

Fix a uniformiser π at P. By (6.3.20) (applied to $D := \partial_\pi$), $\partial_\pi = \partial_\pi(x) \cdot \partial_x$, so $z\,\partial_\pi(t) = (z \cdot \partial_x(t))\,\partial_\pi(x)$. By (6.3.50), for every j,

$$\mathrm{Res}_{P_j'}((z \otimes 1)\,\mathrm{d}(t \otimes 1)) = L_\pi((z\,\partial_\pi(t)) \otimes 1)_{-1} = L_\pi(((z \cdot \partial_x(t))\,\partial_\pi(x)) \otimes 1)_{-1} = \mathrm{Res}_{P_j'}(((z \cdot \partial_x(t)) \otimes 1)\,\mathrm{d}(x \otimes 1))$$

Summing over j (6.3.53) gives $\mathrm{Res}_P(z\,\mathrm{d}t) = \mathrm{Res}_P((z \cdot \partial_x(t))\,\mathrm{d}x)$. ∎

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6.3.9 The Algebraic Residue Theorem

This subsection proves the (algebraic) Residue Theorem for a general base field k, by reduction to the algebraically closed case of (6.3.34).

Proposition 6.3.57 (Algebraic Residue Theorem)
For $t, f \in K$,

$$\sum_P \operatorname{Res}_P(f\,\mathrm{d}t) = 0,$$

summed over all places P of K/k (finitely many terms nonzero, as $f\,\mathrm{d}t$ has only finitely many poles).

Proof. Drafted by Claude Sonnet 5, reviewed by David Rufino.

By (6.3.47) (applied to K/k), restriction $\overline{P} \mapsto \overline{P} \cap K$ is a bijection $\{\text{places of } K/k\} \leftrightarrow \left\{\operatorname{Aut}\left(\overline{k}/k\right)\text{-orbits of places of } K_{\overline{k}}/\overline{k}\right\}$ (where $K_{\overline{k}} := K \otimes_k \overline{k}$, a field by (3.36.45) as in (6.3.43)); the fibre over P is $\left\{P'_1, \ldots, P'_{d_P}\right\}$, $d_P := \deg P$. By (6.3.53), for every place P of K/k and $z \in K$,

$$\operatorname{Res}_P(z\,\mathrm{d}t) = \sum_{j=1}^{d_P} \operatorname{Res}_{P'_j}((z \otimes 1)\,\mathrm{d}(t \otimes 1)).$$

Summing over all places P of K/k and using that the fibres $\left\{P'_1, \ldots, P'_{d_P}\right\}$ partition the places of $K_{\overline{k}}/\overline{k}$ (restriction being a surjection),

$$\sum_{P \text{ of } K/k} \operatorname{Res}_P(f\,\mathrm{d}t) = \sum_{\overline{P} \text{ of } K_{\overline{k}}/\overline{k}} \operatorname{Res}_{\overline{P}}((f \otimes 1)\,\mathrm{d}(t \otimes 1)) = 0$$

by (6.3.34) (applicable as $K_{\overline{k}}/\overline{k}$ is itself an algebraic function field, over the algebraically closed field $\overline{k}$, (6.3.43)), applied to $K_{\overline{k}}/\overline{k}$, $t \otimes 1$, $f \otimes 1$. ∎

6.3.10 Differentials are Weil Differentials, for General k

Fix a transcendental $t \in K$. By (6.3.20), every $D \in \operatorname{Der}_k(K, K)$ is $D(t) \cdot \partial_t$, so a differential $z\,\mathrm{d}t$ ($z \in K$) already carries all the information of an element of the (not yet formally constructed) module $\Omega_{K/k}$, and $\operatorname{Res}_P(z\,\mathrm{d}t)$ is available at every place (6.3.53). We show this recovers Ω_K exactly, generalising (6.3.19) beyond $k = \mathbb{C}$: the complex-analytic ingredients there (contour integration, Stokes' theorem, Mittag-Leffler) are replaced by the algebraic Residue Theorem, the trace formalism of (6.3.53), and the following fact, taken as given in this subsection.

Proposition 6.3.58 (Ω_K is a K-Vector Space)
Let K/k be an algebraic function field. For $x \in K$ and $\omega \in \Omega_K(D)$ ($D \in \operatorname{Div}(K)$), define $x\omega \in \operatorname{Hom}_k(\mathcal{A}_K, k)$ by $(x\omega)(\alpha) := \omega(x\alpha)$ (using that $\mathcal{A}_K$ is a K-algebra). Then $x\omega \in \Omega_K(D - (x)) \subset \Omega_K$, and this action makes Ω_K a K-vector space.

Proof. Drafted by Claude Sonnet 5, reviewed by David Rufino.

$x\omega \in \Omega_K(D - (x))$. For $\alpha \in \mathcal{A}_K(D - (x))$, $v_P(\alpha_P) \geq -v_P(D - (x)) = v_P(x) - v_P(D)$ for every P, so $v_P(x\alpha_P) \geq -v_P(D)$, that is $x\alpha \in \mathcal{A}_K(D)$; and $xK = K$. As $\omega \in \Omega_K(D)$ vanishes on $\mathcal{A}_K(D) + K$, it follows that $(x\omega)(\alpha) = \omega(x\alpha) = 0$ for every $\alpha \in \mathcal{A}_K(D - (x)) + K$, that is $x\omega \in \Omega_K(D - (x))$.

Module axioms. For $x, y \in K$, $\lambda \in k$, $\alpha, \beta \in \mathcal{A}_K$: $(x\omega)(\alpha + \beta) = \omega(x\alpha + x\beta) = (x\omega)(\alpha) + (x\omega)(\beta)$ and $(x\omega)(\lambda\alpha) = \omega(\lambda x\alpha) = \lambda(x\omega)(\alpha)$, so $x\omega \in \operatorname{Hom}_k(\mathcal{A}_K, k)$ is again k-linear; $(x(y\omega))(\alpha) = (y\omega)(x\alpha) = \omega(yx\alpha) = ((xy)\omega)(\alpha)$ and $1\omega = \omega$; $((x + y)\omega)(\alpha) = \omega((x + y)\alpha) = \omega(x\alpha) + \omega(y\alpha) = (x\omega)(\alpha) + (y\omega)(\alpha)$; and $(\omega \mapsto x\omega)$ is k-linear by the same computation applied to $\omega_1 + \omega_2$. These are exactly the axioms making Ω_K a K-module, that is a K-vector space (being already a k-vector space). ∎

Proposition 6.3.59 (Ω_K is 1-Dimensional over K)
Let K/k be an algebraic function field. Then $\dim_K \Omega_K = 1$ (for the K-vector space structure of (6.3.58)).

Proposition 6.3.60 (Differentials are Weil Differentials)
Let K/k be an algebraic function field and $t \in K$ transcendental. For $z \in K$, define $\varphi_{z\,\mathrm{d}t} \in \operatorname{Hom}_k(\mathcal{A}_K, k)$ by

$$\varphi_{z\,\mathrm{d}t}(\alpha) := \sum_P \operatorname{Res}_P(\alpha_P z\,\mathrm{d}t), \qquad \alpha = (\alpha_P)_P \in \mathcal{A}_K$$

(a finite sum: $\alpha_P \in \mathcal{O}_P$ for all but finitely many P, and $z\,\mathrm{d}t$ is regular at all but finitely many P, so $\operatorname{Res}_P(\alpha_P z\,\mathrm{d}t) = 0$ for almost all P (6.3.55)). Then $\varphi_{z\,\mathrm{d}t} \in \Omega_K$, and

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$$K \longrightarrow \Omega_K, \qquad z \longmapsto \varphi_{z\,\mathrm{d}t}$$

is a canonical K-linear isomorphism (for the K-vector space structures of (6.3.58) and K itself).

Proof. Drafted by Claude Sonnet 5, reviewed by David Rufino.

$\varphi_{z\,\mathrm{d}t} \in \Omega_K$. Let D be the divisor with $v_P(D) := v_P(z\,\partial_\pi(t))$ for a uniformiser π at P (independent of the choice of π: for another uniformiser π', $\partial_{\pi'}(t) = \partial_{\pi'}(\pi)\,\partial_\pi(t)$ (6.3.22), with $\partial_{\pi'}(\pi)$ a unit at P, so $v_P(z\partial_{\pi'}(t)) = v_P(z\partial_\pi(t))$; and this vanishes for all but finitely many P, as z has finitely many poles and t finitely many poles). For $\alpha \in \mathcal{A}_K(D)$, $v_P(\alpha_P) \geq -v_P(D) = -v_P(z\partial_\pi(t))$ for every P, so $v_P(\alpha_P z\partial_\pi(t)) \geq 0$, that is $\alpha_P z\,\partial_\pi(t) \in \mathcal{O}_P$, giving $\mathrm{Res}_P(\alpha_P z\,\mathrm{d}t) = 0$ (6.3.55) so $\varphi_{z\,\mathrm{d}t}(\alpha) = 0$. For $f \in K$ embedded diagonally, $\varphi_{z\,\mathrm{d}t}(f, f, \ldots) = \sum_P \mathrm{Res}_P(fz\,\mathrm{d}t) = 0$ by (6.3.57) (the Residue Theorem), applied to $fz \in K$. So $\varphi_{z\,\mathrm{d}t}$ vanishes on $\mathcal{A}_K(D) + K$, that is $\varphi_{z\,\mathrm{d}t} \in \Omega_K(D) \subset \Omega_K$.

***K*-linearity.** k-linearity in z is immediate from (6.3.54).a), applied place by place and summed. For $x \in K$, using $x\alpha_P \cdot z = \alpha_P \cdot (xz)$ in the field K,

$$(x\,\varphi_{z\,\mathrm{d}t})(\alpha) = \varphi_{z\,\mathrm{d}t}(x\alpha) = \sum_P \mathrm{Res}_P((x\alpha_P)z\,\mathrm{d}t) = \sum_P \mathrm{Res}_P(\alpha_P(xz)\,\mathrm{d}t) = \varphi_{(xz)\,\mathrm{d}t}(\alpha),$$

so $x\,\varphi_{z\,\mathrm{d}t} = \varphi_{(xz)\,\mathrm{d}t}$: the map is K-linear (6.3.58).

Injectivity. Suppose $\varphi_{z\,\mathrm{d}t} = 0$ and $z \neq 0$; we derive a contradiction. Fix a place P and a uniformiser π at P, and let $d, P'_1, \ldots, P'_d, \iota_1, \ldots, \iota_d$ be as in (6.3.49). As ∂_π is a nonzero derivation and t is transcendental, $\partial_\pi(t) \neq 0$ (6.3.20), so $w := z\,\partial_\pi(t) \neq 0$ in K. Since $w \in K$ is fixed pointwise by $\mathrm{Aut}\big(\overline{k}/k\big)$ (embedded diagonally in $K_{\overline{k}}$) while σ_* permutes $\{P'_1, \ldots, P'_d\}$ preserving valuations (6.3.47), $m := v_{P'_j}(w \otimes 1)$ is independent of j. Let $c := \pi^{-m-1} \in K^\star$; then $v_{P'_j}(cw \otimes 1) = -1$ for every j, so

$$u_j := \mathrm{Res}_{P'_j}((cz \otimes 1)\,\mathrm{d}(t \otimes 1))$$

is the (nonzero) leading Laurent coefficient of $cw \otimes 1$ at P'_j. By (6.3.52) (applied to t, cz in place of t, z), $u_j = \iota_j(\gamma)$ for a single $\gamma \in k(\mathfrak{m}_P)$; as $u_1 \neq 0$ and ι_1 is injective, $\gamma \neq 0$. As k is perfect, $k(\mathfrak{m}_P)/k$ is separable (3.19.103), so by (3.19.131) there is $\overline{e} \in k(\mathfrak{m}_P)$ with $\mathrm{Tr}_{k(\mathfrak{m}_P)/k}(\overline{e}\gamma) \neq 0$; lift it to a unit $e \in \mathcal{O}_P^\star$ (any lift of a nonzero residue class is a unit, $\mathcal{O}_P$ being local). As $e \otimes 1$ is a unit of $\mathcal{O}_{P'_j}$ with residue class $\iota_j(\overline{e})$, and $cw \otimes 1$ has valuation exactly -1 with leading coefficient $u_j = \iota_j(\gamma)$, the product $(e \otimes 1)(cw \otimes 1)$ again has valuation exactly -1 (no lower-order term of $e \otimes 1$ to interact with, as $cw \otimes 1$ has none below X^{-1}), with leading coefficient $\iota_j(\overline{e})\,\iota_j(\gamma) = \iota_j(\overline{e}\gamma)$: that is $\mathrm{Res}_{P'_j}((ecz \otimes 1)\,\mathrm{d}(t \otimes 1)) = \iota_j(\overline{e}\gamma)$ for every j, so

$$\mathrm{Res}_P(ecz\,\mathrm{d}t) = \sum_{j=1}^{d} \iota_j(\overline{e}\gamma) = \mathrm{Tr}_{k(\mathfrak{m}_P)/k}(\overline{e}\gamma) \neq 0.$$

Testing $\varphi_{z\,\mathrm{d}t}$ against the repartition supported at P with value $ec \in K$ then gives $\varphi_{z\,\mathrm{d}t}(\ldots) = \mathrm{Res}_P(ecz\,\mathrm{d}t) \neq 0$, contradicting $\varphi_{z\,\mathrm{d}t} = 0$.

Surjectivity. The map $K \to \Omega_K$ is K-linear and injective, and K is a 1-dimensional K-vector space over itself; so its image is a 1-dimensional K-subspace of Ω_K, which by (6.3.59) is itself 1-dimensional over K. So the image is all of Ω_K. ∎

6.3.11 Tate's Approach to Residues: A Sketch

The route taken above defines Res_P through a uniformiser and the formal derivative, and proves the Residue Theorem (6.3.57) by reducing to $K = k(x)$ along a finite separable extension. Tate [Tat68] gives a different, intrinsic route, which we only sketch here (nothing later depends on it): the residue is defined from linear algebra alone, and the Residue Theorem is deduced from the finiteness results of the Riemann-Roch theory.

a) *Traces of finite-rank operators.* The trace of a finite-rank endomorphism T of a k-vector space V is its trace on any finite-dimensional subspace containing $T(V)$. It is additive, $\mathrm{Tr}(AB) = \mathrm{Tr}(BA)$ whenever $A : V \to W$, $B : W \to V$ and one of them has finite rank, and $\mathrm{Tr}(T) = \mathrm{Tr}(T|_H) + \mathrm{Tr}\big(\overline{T}\big)$ if T preserves a finite-dimensional H, with $\overline{T}$ the induced operator on V/H. In particular $\mathrm{Tr}([T, F]) = 0$ for F of finite rank and T arbitrary.

b) *The local residue.* Let π be a k-linear retraction of the local space onto the lattice $\mathcal{O}_P$ (in Tate's setting the completion K_P, with π the projection killing the principal parts). For f in the local space let $T_f := \pi \circ (f \cdot -)|_{\mathcal{O}_P}$ be the Toeplitz operator on $\mathcal{O}_P$. Then $\big[T_f, T_g\big]$ has finite rank, because $(1 - \pi)(f\mathcal{O}_P)$ lies in the finite-dimensional image of $t^{-n}\mathcal{O}_P/\mathcal{O}_P$, and $\mathrm{Res}_P(f\,\mathrm{d}g) := \mathrm{Tr}\big(\big[T_f, T_g\big]\big)$. Changing π changes each T_f by a finite-rank operator, which does not change the trace of the commutator. So no uniformiser, derivation or chain rule enters.

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c) *Local properties.* It is k-linear, vanishes on differentials regular at P, has $\operatorname{Res}_P(\mathrm{d}t/t) = \deg P$ (the trace is taken over k) and is invariant under automorphisms of K_P; it should agree with the trace of the coefficient of t^{-1} used above.

d) *Global input.* The finiteness underlying Riemann-Roch: $H \coloneqq K \cap \mathcal{A}_K(0) = L(0)$ and $\mathcal{A}_K/(K + \mathcal{A}_K(0))$ are finite dimensional (6.3.15) and $\mathcal{A}_K(D)/\mathcal{A}_K(0) = \bigoplus_P t_P^{-n_P}\mathcal{O}_P/\mathcal{O}_P$ is finite dimensional for every divisor $D = \sum_P n_P P$.

e) *The Residue Theorem as a trace identity.* Write $\mathcal{A} \coloneqq \mathcal{A}_K$ and $L \coloneqq \mathcal{A}(0) = \prod_P \mathcal{O}_P$, and for $f, g \in K$ let D be a divisor with $f\mathcal{A}(0), g\mathcal{A}(0) \subset \mathcal{A}(D)$.

   (a) For the retraction $\pi_L \coloneqq \prod_P \pi_P : \mathcal{A} \to L$ and $T_f \coloneqq \pi_L \circ (f \cdot -)|_L$, the commutator $\big[T_f, T_g\big]$ is the direct sum over P of the local commutators, of which only finitely many are nonzero (where f, g are regular the operators are multiplication by f, g and commute). So $\operatorname{Tr}\big(\big[T_f, T_g\big]\big) = \sum_P \operatorname{Res}_P(f\,\mathrm{d}g)$.

   (b) As in b), $\operatorname{Tr}\big(\big[T_f, T_g\big]\big)$ does not depend on the retraction $\pi : \mathcal{A} \to L$: another retraction changes T_f by $\psi \circ (1-\pi) f|_L$, which has finite rank since $(1-\pi)(\mathcal{A}(D))$ is finite dimensional.

   (c) Choose π adapted to K: write $K = H \oplus W_K$ and choose a finite-dimensional F with $\mathcal{A} = L \oplus W_K \oplus F$ (possible by d)), and let π be the projection onto L along $W_K \oplus F$. For $h \in H$, $fh \in K = H \oplus W_K$ so $T_f h \in H$: both T_f, T_g preserve H and act there as operators on a finite-dimensional space, whose commutator has trace 0.

   (d) On $L/H \cong (L+K)/K =: \Lambda \subset \mathcal{A}/K$ the induced operator $\overline{T}_f$ is the Toeplitz operator of multiplication by f on $\mathcal{A}/K$ (where f, g commute exactly), with respect to the retraction $\mathcal{A}/K = \Lambda \oplus \overline{F} \to \Lambda$. Writing $\overline{\pi} = 1 - p$ with p the projection onto $\overline{F}$, and using $fg = gf$ on $\mathcal{A}/K$, one finds $\big[\overline{T}_f, \overline{T}_g\big] = -\overline{\pi} f p g + \overline{\pi} g p f$ on Λ, whose trace equals $\operatorname{Tr}_{\overline{F}}([pg|_{\overline{F}}, pf|_{\overline{F}}]) = 0$, a commutator of operators on the finite-dimensional space $\overline{F}$.

   By (c), (d) and additivity of the trace, $\operatorname{Tr}\big(\big[T_f, T_g\big]\big) = 0$, so $\sum_P \operatorname{Res}_P(f\,\mathrm{d}g) = 0$ by (a) and (b).

Nothing here uses the topology of K_P. The construction of b) uses only the subspace $\mathcal{O}_P \subset K$, a retraction $\pi : K \to \mathcal{O}_P$, and $\dim_k t^{-n}\mathcal{O}_P/\mathcal{O}_P < \infty$ (as $k(\mathfrak{m}_P)/k$ is finite): "lattices" and "continuity" reduce to commensurability of subspaces and to the pole order of f, so the local residue may be defined on K itself, and the global step on the repartitions $\mathcal{A}_K$ of this book (components in K, not in the K_P) together with the finiteness of d). Compared with the proof above, this avoids the reduction to $k(x)$, the trace formula and separability, and works in any characteristic. What is left to check is that this Res_P agrees with the Laurent-coefficient residue used above (for a place of degree d, $\operatorname{Res}_P(\mathrm{d}t/t) = d$ with the trace taken over k); the global argument of e) is then complete.

6.3.12 Sheaf-Theoretic Interpretation of Repartitions

Unlike the rest of this chapter, this section depends on sheaf theory developed later in this book (4.2.2), and so is not logically self-contained at this point in the text; nothing later in this chapter depends on it.

Let $X \coloneqq \mathbb{P}_K$, the set of places of K/k, equipped with the **cofinite topology** (opens are $\varnothing$ and complements of finite subsets); as $\mathbb{P}_K$ is infinite, any two nonempty opens intersect, so X is irreducible. Let $\mathcal{K}_X$ be the constant sheaf with value K (4.2.50). For $D \in \operatorname{Div}(K)$ define the subsheaf

$$\mathcal{O}(D)(U) \coloneqq \{x \in K \mid v_P(x) \geq -v_P(D)\ \forall P \in U\} \subseteq \mathcal{K}_X(U)$$

generalising the structure sheaf ($\mathcal{O}(0) = \mathcal{O}_X$), with inclusion $\iota_D : \mathcal{O}(D) \hookrightarrow \mathcal{K}_X$.

Proposition 6.3.61 (Stalk of $\mathcal{O}(D)$)
For $P \in X$, the canonical map $\mathcal{O}(D)_P \to \mathcal{K}_{X,P} = K$ is injective, identifying $\mathcal{O}(D)_P$ with the fractional ideal

$$\{y \in K \mid v_P(y) \geq -v_P(D)\} =: \pi_P^{-v_P(D)}\mathcal{O}_P$$

of K.

Proof. Drafted by Claude Sonnet 5, reviewed by David Rufino.

Stalk of $\mathcal{K}_X$. Since X is irreducible, $\mathcal{K}_X(U) = K$ for every nonempty open $U \subset X$, with every restriction map the identity (4.2.50). We can therefore represent the stalk as $\mathcal{K}_{X,P} = K$ with the identity maps. Under this identification, the canonical map $\mathcal{O}(D)_P \to \mathcal{K}_{X,P} = K$ is given by $(U, x) \mapsto x$.

Injectivity. Two representatives $(U, x), (U', x')$ of elements of $\mathcal{O}(D)_P$ become identified in the stalk exactly when they agree on some common refinement; as restriction maps are the identity, this means $x = x'$ already, as elements of K. So $(U, x) \mapsto x$ is injective, and we identify $\mathcal{O}(D)_P$ with its image in K.

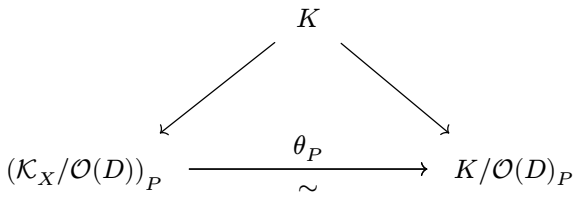
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$\mathcal{O}(D)_P \subseteq \{y : v_P(y) \geq -v_P(D)\}$. Every element of $\mathcal{O}(D)_P$ is represented by a pair (U, x), $P \in U$, $x \in \mathcal{O}(D)(U)$, whose image in $\mathcal{O}(D)_P$ is x itself (viewed in K); taking the defining condition of $\mathcal{O}(D)(U)$ at the point $P \in U$ gives $v_P(x) \geq -v_P(D)$.

$\{y : v_P(y) \geq -v_P(D)\} \subseteq \mathcal{O}(D)_P$. Let $y \in K$ with $v_P(y) \geq -v_P(D)$. By (6.2.116), y has finitely many poles, so together with finiteness of $\mathrm{Supp}(D)$, the set $T := \{Q \in X \mid v_Q(y) < -v_Q(D)\}$ is finite. As $P \notin T$, $U := X \setminus T$ is an open neighbourhood of P, and $y \in \mathcal{O}(D)(U)$ by construction of T; so (U, y) represents an element of $\mathcal{O}(D)_P$ mapping to y.

Fractional ideal. By the previous two paragraphs $\mathcal{O}(D)_P = \{y \in K \mid v_P(y) \geq -v_P(D)\}$. Writing π_P for a uniformiser of $\mathcal{O}_P$, (3.31.2) represents every $y \in K^\star$ uniquely as $y = u\pi_P^{v_P(y)}$ for $u \in \mathcal{O}_P^\star$, so $v_P(y) \geq -v_P(D) \iff y \in \pi_P^{-v_P(D)}\mathcal{O}_P$; thus $\mathcal{O}(D)_P = \pi_P^{-v_P(D)}\mathcal{O}_P$. This is the image of $\mathcal{O}_P$ under multiplication by the fixed element $\pi_P^{-v_P(D)} \in K$, hence an $\mathcal{O}_P$-submodule of K and so a fractional ideal of $\mathcal{O}_P$ (3.30.1) (indeed $\mathcal{O}_P$ is a discrete valuation ring by (6.2.114), so this is the fractional ideal $\mathfrak{m}_P^{-v_P(D)}$ of (3.31.7)); in particular it is a k-vector subspace of K, since $k \subset \mathcal{O}_P$ (6.2.113). ∎

Proposition 6.3.62 (Stalk of the Cokernel)
Since ι_D is a monomorphism (6.3.61), write $\mathcal{K}_X/\mathcal{O}(D) := \mathrm{coker}(\iota_D)$ (4.2.36). There is a unique morphism θ_P making



commute, and θ_P is (moreover) an isomorphism; in particular $(\mathcal{K}_X/\mathcal{O}(D))_P \cong K/\mathcal{O}(D)_P = K/\pi_P^{-v_P(D)}\mathcal{O}_P$.

Proof. Drafted by Claude Sonnet 5, reviewed by David Rufino.

This is (4.2.37), applied to $\iota_D : \mathcal{O}(D) \to \mathcal{K}_X$ (using $\mathcal{K}_{X,P} = K$), together with (6.3.61) identifying $\mathcal{O}(D)_P$ with $\pi_P^{-v_P(D)}\mathcal{O}_P$. ∎

Used freely below.

Proposition 6.3.63 (Repartitions compute the Cokernel Sheaf)
Writing $p_X : K = \Gamma(X, \mathcal{K}_X) \to \Gamma(X, \mathcal{K}_X/\mathcal{O}(D))$ for the global sections of the cokernel map (4.2.35), there is an exact sequence of k-vector spaces

$$0 \longrightarrow L(D) \longrightarrow K \xrightarrow{p_X} \Gamma(X, \mathcal{K}_X/\mathcal{O}(D)) \xrightarrow{\Phi} \mathcal{A}_K/(\mathcal{A}_K(D) + K) \longrightarrow 0$$

where Φ sends $\tau \in \Gamma(X, \mathcal{K}_X/\mathcal{O}(D))$ – a function $\tau : X \to \coprod_{P \in X} K/\mathcal{O}(D)_P$ (4.2.35) – to the class of any repartition $(\alpha_P)_P \in \mathcal{A}_K$ with $\alpha_P \equiv \tau(P) \pmod{\mathcal{O}(D)_P}$ for every P; that is, sending a representative $x \in K$ to its corresponding residue vector. In particular $i(D)$ is (via (6.3.63)) the dimension of the cokernel of p_X – via sheaf cohomology, the group $H^1(X, \mathcal{O}(D))$, a cohomological invariant of X, not merely of the algebra of repartitions.

Sketch. $(\iota_D)_U$ is the inclusion $\mathcal{O}(D)(U) \hookrightarrow K$, injective.

By (4.2.35) a global section of $\mathcal{K}_X/\mathcal{O}(D)$ is represented by a cover of X together with representatives in K over each open, and since the cofinite topology is quasi-compact (4.1.93) this cover may be taken finite, say $X = U_1 \cup \cdots \cup U_n$ together with representatives $x_i \in K$, well-defined modulo $\mathcal{O}(D)(U_i)$, that **agree at every point of overlap**: for every place $P \in U_i \cap U_j$, the images of x_i, x_j in the stalk $(\mathcal{K}_X/\mathcal{O}(D))_P = K/\pi_P^{-v_P(D)}\mathcal{O}_P$ (6.3.62) coincide, that is

$$v_P(x_i - x_j) \geq -v_P(D)$$

Given such data, set $\alpha_P := x_i$ (a literal element of K, well-defined only up to the choice of i) for any i with $P \in U_i$. The class of α_P modulo $\pi_P^{-v_P(D)}\mathcal{O}_P$ is independent of the choice of i: if also $P \in U_j$, then $P \in U_i \cap U_j$, so the overlap condition above applies directly at this P and gives $v_P(x_i - x_j) \geq -v_P(D)$, that is $x_i \equiv x_j \pmod{\pi_P^{-v_P(D)}\mathcal{O}_P}$. Each x_i has finitely many poles (6.2.116), so $\{P \mid \alpha_P \notin \mathcal{O}_P\} \subseteq \bigcup_{i=1}^n \{P \in U_i \mid v_P(x_i) < 0\}$, a finite union of finite sets, is finite; so $\alpha_P \in \mathcal{O}_P$ for all but finitely many P, giving $(\alpha_P)_P \in \mathcal{A}_K$.

Independence of the choice of representing data. Let $(U_i, x_i)_{i=1}^n$ and $(U_i', x_i')_{i=1}^m$ be two choices representing the same τ, with resulting repartitions $(\alpha_P)_P, (\alpha_P')_P \in \mathcal{A}_K$ as above. For each P, choosing i, i' with $P \in U_i \cap U_i'$, both x_i and x_i' represent $\tau(P) \in K/\mathcal{O}(D)_P$ (4.2.35), that is $x_i \equiv x_i' \pmod{\mathcal{O}(D)_P}$, so $\alpha_P - \alpha_P' = x_i - x_i' \in \mathcal{O}(D)_P$.

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As this holds for every P, $(\alpha_P - \alpha'_P)_P \in \mathcal{A}_K(D)$; so $(\alpha_P)_P$ and $(\alpha'_P)_P$ have the same class in $\mathcal{A}_K/(\mathcal{A}_K(D)+K)$. This assignment therefore descends to a well-defined map

$$\Phi : \Gamma(X, \mathcal{K}_X/\mathcal{O}(D)) \longrightarrow \mathcal{A}_K/(\mathcal{A}_K(D)+K)$$

Exactness at K. For $c \in K$, $p_X(c)$ is represented by the trivial cover $\{X\}$ with $x_1 := c$, so $p_X(c) = 0$ if and only if $c \in \mathcal{O}(D)(X) = L(D)$; that is, $\ker(p_X) = L(D)$, matching exactness at K (the map $L(D) \hookrightarrow K$ being the inclusion).

Exactness at $\Gamma(X, \mathcal{K}_X/\mathcal{O}(D))$. ($\mathrm{Im}(p_X) \subseteq \ker(\Phi)$) For $c \in K$, $p_X(c)$ has $\alpha_P = c$ for every P, that is, the constant repartition $\delta(c)$ (diagonal embedding $\delta : K \hookrightarrow \mathcal{A}_K$), whose class in $\mathcal{A}_K/(\mathcal{A}_K(D)+K)$ is 0, since $\delta(c) \in K$; so $\Phi(p_X(c)) = 0$.

($\ker(\Phi) \subseteq \mathrm{Im}(p_X)$) Suppose $\Phi(\tau) = 0$, that is $(\alpha_P)_P \in \mathcal{A}_K(D) + K$, say $(\alpha_P)_P = \beta + \delta(c)$ with $\beta \in \mathcal{A}_K(D)$, $c \in K$. Then $x_i - c \in \mathcal{O}(D)(U_i)$ for every i (as $x_i - c = \alpha_P - c = \beta_P \in \mathcal{O}(D)_P$ for every $P \in U_i$), so τ and $p_X(c)$ are represented by the same local data, that is $\tau = p_X(c) \in \mathrm{Im}(p_X)$.

Exactness at $\mathcal{A}_K/(\mathcal{A}_K(D)+K)$. Given $(\alpha_Q)_Q \in \mathcal{A}_K$, let

$$S := \{Q \mid \alpha_Q \notin \mathcal{O}_Q\} \cup \mathrm{Supp}(D) = \{P_1, ..., P_n\}$$

finite, being a union of two finite sets; note $v_Q(D) = 0$ for every $Q \notin S$, so $\mathcal{O}(D)_Q = \mathcal{O}_Q$ there. Set $x_0 := 0$ and, for $i = 1, ..., n$, $x_i := \alpha_{P_i}$ and $Z_i := \{Q \neq P_i \mid \alpha_{P_i} \notin \mathcal{O}_Q\}$, the poles of α_{P_i} other than possibly at P_i itself, finite by (6.2.116). Since the cofinite topology has no "small" neighbourhoods – any nonempty open already excludes only finitely many points – each U_i ($i \geq 1$) must be taken to also avoid x_i's own poles, not merely to contain P_i: set

$$U_0 := X \setminus S, \qquad U_i := X \setminus ((S \cup Z_i) \setminus \{P_i\}) \quad (i = 1, ..., n)$$

These cover X: $U_0 \cup \cdots \cup U_n \supseteq (X \setminus S) \cup \{P_1, ..., P_n\} = X$, as $P_i \in U_i$ for each i.

Each U_i meets S in at most the single point P_i: $U_0 \cap S = \varnothing$, and $U_i \cap S = \{P_i\}$ for $i \geq 1$, since U_i excludes every other point of S. Consequently, for $i \neq j$, no point of S lies in both U_i and U_j (it would have to be the marked point of each, and the marked points – or absence thereof, for $i = 0$ – differ), so $U_i \cap U_j \subseteq X \setminus S$; in particular $Q \in U_i \cap U_j$ gives $v_Q(D) = 0$ and $Q \neq P_i, P_j$.

They agree at overlaps: for $i \neq j$ in $\{0, ..., n\}$ and $Q \in U_i \cap U_j$, $v_Q(x_i) \geq 0$ (trivially if $i = 0$, as $x_0 = 0$; else since $Q \in U_i$, $Q \neq P_i$ gives $Q \notin Z_i$) and likewise $v_Q(x_j) \geq 0$; so by the ultrametric inequality $v_Q(x_i - x_j) \geq \min(v_Q(x_i), v_Q(x_j)) \geq 0 = -v_Q(D)$.

This data therefore glues to a class $\tau \in \Gamma(X, \mathcal{K}_X/\mathcal{O}(D))$ with $\tau(Q) = \alpha_Q \pmod{\mathcal{O}(D)_Q}$ for every Q (on U_0 this matches $x_0 = 0$, since $Q \notin S$ gives $\alpha_Q \in \mathcal{O}_Q = \mathcal{O}(D)_Q$; at $Q = P_i$ this matches $x_i = \alpha_{P_i}$ directly), so $\Phi(\tau) = (\alpha_Q)_Q$ exactly; so Φ is surjective, matching exactness at the final term. ∎

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